IE2106 Engineering Mathematics I

Probability & Statistics

by

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A. Introduction

Statistics is the science of collecting, presenting, analyzing and interpreting data for making decisions under conditions of uncertainty.

Statistics plays a key role in many activities, e.g.

- **Performing the quality control techniques**
- **Getting a new drug approved by the Food and Drug Administration**

- **Predicting the outcome of elections**

- **Conducting audits of banks, government agencies, and other businesses**

The foundation of statistics is *probability theory*. Probability theory is the mathematical modeling of the phenomenon of chance or randomness. It deals with averages of unpredictable phenomena. The first ideas took shape at gambling tables.

1. Set Theory

- A *set* is a collection of objects. We usually use capital letters to denote sets, e.g., A is the set of females in this room.

- The members of a set A are called the *elements* of A. For e.g.,

  Patricia is an element of A (Patricia $\in A$)

  Patrick is not an element of A (Patrick $\notin A$).

- The *universal set*, U, is the set of all objects under consideration. For e.g., U is the set of all people in this room.

- The *null set* or *empty set*, $\emptyset$, has no elements. For e.g., the set of females above 1.8m tall in this room is an empty set.

- Let A and B be 2 sets. A is a *subset* of B if each element of A is also an element of B. Write $A \subset B$. For e.g.,

  the set of females wearing metal frame glasses in this room
  $\subset$ the set of females wearing glasses in this room
  $\subset$ the set of females in this room.

Example 1.1.

(a) The set of females in this room is

$$A = \{Patricia,\ Jenny,\ \cdots,\ Girl\ Girl\}.$$

(b) The set of integers less than 10 and greater than 5 is

$$A = \{6, 7, 8, 9\}.$$

(c) The set of real numbers between 0 and 1 inclusive is

$$A = \{x\ :\ x \in \mathbb{R},\ 0 \leq x \leq 1\}.$$

1.1. Some Operations on Sets

Let A and B be any subsets of the universal set U.

- The *complement* of A (w.r.t. U), A', is the set made up of the elements of U that do not belong to A, i.e.,

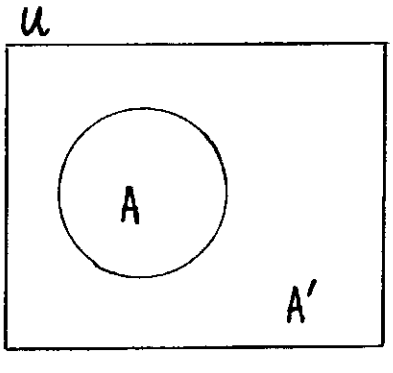
$$A' = \{x \ : \ x \in U, \ x \notin A\} = U \backslash A.$$

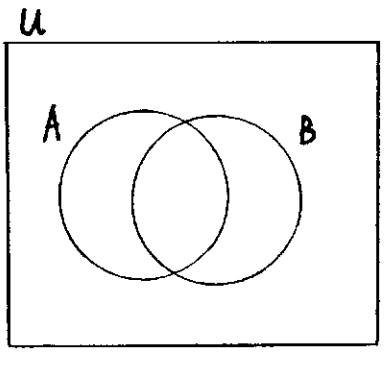
- The *intersection* of A and B, $A \cap B$, is the set of elements that belong to both A and B, i.e.,

$$A \cap B = \{x \ : \ x \in A \text{ and } x \in B\}.$$

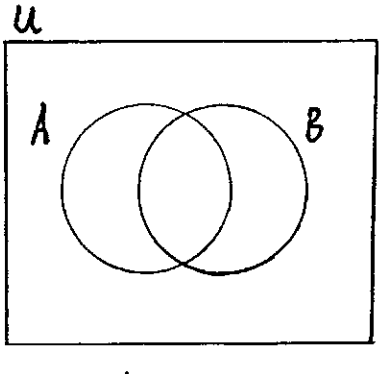
- The *union* of A and B, $A \cup B$, is the set of elements that belong to at least one of the sets A and B, i.e.,

$$A \cup B = \{x \ : \ x \in A \text{ or } x \in B \text{ (or both) }\}.$$





$A \cap B$



$A \cup B$

Example 1.2. Let the universal set be

$$U = \{1, 2, 3, 4, 5, 6, 7\},$$

and A, B, C be 3 subsets

$$A = \{1, 2, 3\}, \qquad B = \{2, 4, 6\}, \qquad C = \{1, 3, 5, 7\}.$$

Then,

$$A \cup (B \cap C') = A \cup (B \cap \{2, 4, 6\}) = A \cup B = \{1, 2, 3, 4, 6\}$$

$$A' \cup (B \cap C) = A' \cup \emptyset = A' = \{4, 5, 6, 7\}$$

$$(A \cup B) \cap (B \cap C) = \{1, 2, 3, 4, 6\} \cap \emptyset = \emptyset$$

B. Probability

1. Experiments, Sample Spaces, Events

1.1. Experiments & Sample Spaces

- In the study of probability, any process of observation is referred to as an *experiment*.

- The results of an observation are called the *outcomes* of the experiment.

- **A *random experiment* is an experiment in which the outcome varies in an unpredictable fashion when the experiment is repeated under the same conditions. For e.g., if a die is tossed, it is not certain that, say, a '6' will appear.**

  - **Although we cannot predict the outcomes with certainty, we can describe the set of possible outcomes.**
  - **The experiment can be repeated under the same conditions. As the no. of repetitions increases, certain patterns in the frequency of outcome occurrence emerge.**

- The *sample space* S of a random experiment is defined as the set of all possible outcomes. A particular outcome, i.e., an element of S, is called a *sample point*.

- A *discrete* sample space is one which has a finite no. of outcomes or countable infinite no. of outcomes. A *continuous* sample space is one which has noncountable outcomes.

Example 1.1.

E1: Toss a coin and note the outcome.

E2: Toss a coin 3 times and note the sequence of heads and tails.

E3: Toss a coin 3 times and note the number of heads.

Denote a head by h, and a tail by t. The sample spaces corresponding to E1 - E3 are

$$S1 = \{h, t\}, S2 = \{hhh,\ hht,\ hth,\ thh,\ tth,\ tht,\ htt,\ ttt\},$$

$S3 = \{0, 1, 2, 3\}$.

Example 1.2. **The electronic device at an ERP gantry is counting the number of vehicles that pass through between 7.30am and 9.30am.**

The sample space is

$S = \{0, 1, 2, 3, \cdots\}$.

Example 1.3. **A new bulb is put on life test. The elapsed time (in hours) at failure is recorded.**

The sample space is

$$S = \{t \ : \ t \in \mathbb{R}, \ t \geq 0\}.$$

1.2. Events

An *event* is a subset of the sample space S. An event is 'something that happens'.

Example 1.4. Refer to Examples 1.1 - 1.3.

(a) E1: $S1 = \{h, t\}$ has $2^2 = 4$ subsets. The following 4 sets are all the events in experiment E1:

$$\{h, t\},\ \{h\},\ \{t\},\ \emptyset.$$

The first is $S = \{h, t\}$, the sample space, and this event occurs if either a head or a tail is obtained. It is a *certain event*.

The null event $\emptyset$, which contains no outcomes, *never occurs*.

(b) **E2: The event** *'the 3 tosses give the same face'* **is** $\{hhh, ttt\}$.

The event *'the 3 tosses give equal no. of heads and tails'* **is** $\emptyset$.

(c) **E3: The event** *'the 3 tosses give odd no. of heads'* **is** $\{1, 3\}$.

The event *'the no. of heads equals 4'* **is** $\emptyset$.

(d) **ERP gantry: The event '*the no. of vehicles recorded is between 1000 and 1500 inclusive*' is**

$$\{k \ : \ k \in \mathbf{Z}, \ 1000 \leq k \leq 1500\}.$$

(e) **Bulb experiment: The event '*the time to failure is greater than 1000 hours*' is**

$$\{t \ : \ t \in \mathbb{R}, \ t > 1000\}.$$

1.3. Complementary Events

The *complement* of any event A is the event that A does not occur. If A is an event contained in a sample space S, then the event '*not* A', denoted by A', is the event containing all the outcomes in S that are not contained in A. So $A' = S \backslash A$.

Example 1.5. Consider the experiment of tossing a six-sided dice. If A is the event of getting a 4 or a 6, then the event A' contains the outcomes 1, 2, 3 and 5, i.e.,

$$A = \{4, 6\}, \qquad A' = \{1, 2, 3, 5\}.$$

Note that $S = A \cup A' = \{1, 2, 3, 4, 5, 6\}$.

1.4. Compound Events

When two or more events are joined by the connectives '*and*' or '*or*', the resulting event is called a *compound event*.

Let A and B be 2 events.

(a) $(A$ or $B) = A \cup B$ is the event that

- **A will occur, or**
- **B will occur, or**
- **they both occur.**

(b) $(A$ and $B) = A \cap B$ is the event that both A and B will occur at the same time.

Example 1.6. Consider the experiment of tossing 2 six-sided dice. Let

A be the event of getting 2 different even numbers,

B be the event of getting 2 numbers that sum up to 6.

$S = \{(i, j) \ : \ i, j = 1, 2, \cdots, 6\}$ (36 sample points)

$A = \{(2, 4), (2, 6), (4, 6), (4, 2), (6, 2), (6, 4)\}$

$B = \{(1, 5), (2, 4), (3, 3), (5, 1), (4, 2)\}$

$(A \text{ and } B) = A \cap B = \{(2, 4), (4, 2)\}$

$(A \text{ or } B) = A \cup B = \{(2,4),(2,6),(4,6),(4,2),(6,2),(6,4),$
$(1,5),(3,3),(5,1)\}$

Summary

1. An *experiment* is a process that leads to one of several possible outcomes. An *outcome* of an experiment is some observation or measurement.

2. An experiment is called a *random experiment* if its outcome cannot be predicted.

3. The *sample space* is the set of all possible outcomes of an experiment. A *sample point* is a particular outcome.

4. An *event* is a subset of a sample space.

5. The *complement* of an event A, A', is the collection of all sample points that are in the sample space but not in A.

6. Let A and B be two events in a sample space. (A or B) $= A \cup B$ is the event containing all the sample points in A or B or both.

7. (A and B) $= A \cap B$ is the event composed of all the sample points that are in both A and B.

1.5. Mutually Exclusive Events

If A and B are events that have no outcomes in common, then A and B are called *mutually exclusive events*, i.e.,

$$(A \text{ and } B) = A \cap B = \emptyset$$

means A and B are mutually exclusive events.

Note also that *mutually exclusive events* cannot happen at the same time!

Example 1.7. Consider the experiment of tossing a coin. '$head$' and '$tail$' are mutually exclusive events because they cannot both occur.

Similarly, when you toss a die, the event of getting a 'i' is mutually exclusive to getting a 'j', $i \neq j, \;\; i, j = 1, 2, \cdots, 6$.

Example 1.8. Consider the experiment of tossing a 10-cent coin, followed by a 20-cent coin. Are the following pairs of events mutually exclusive?

(a) $A =$ two heads, $B =$ two tails

Yes, they have no outcomes in common.

(b) $A = \{ht, tt\},\ B = \{ht, th\}$

No, they have ht in common.

Example *1. Consider the events A and B, are they mutually exclusive events?

$$A = \{\text{the sun is shining brightly}\}$$
$$B = \{\text{it is raining}\}$$

$A = \{\text{the sun is shining brightly}\}$

$B = \{\text{it is snowing}\}$

$$A = \{\textbf{you are sleeping}\}$$

$$B = \{\textbf{you are walking}\}$$

Summary of mutually exclusive events

1. Two events A and B are called *mutually exclusive events* if one of the following holds:

   - A and B have no outcomes in common
   - $(A \text{ and } B) = A \cap B = \emptyset$
   - $P(A \text{ and } B) = 0$
   - A and B cannot happen at the same time

2. If A and B are mutually exclusive events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B).$$

In general, if A and B are any 2 events, not necessarily mutually exclusive, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

2. Basic Probability

Let A be an event and let $P(A)$ denotes the *probability of* A.

- $0 \leq P(A) \leq 1$.
- $P(A)$ indicates the likelihood of occurrence of event A.
- If $P(A) = 0$, then event A is certain not to occur. If $P(A) = 1$, the event is certain to occur. The closer $P(A)$ is to 1, the more likely it is to occur. The closer $P(A)$ is to

0, the more unlikely it is to occur.

2.1. Relative Frequency Approach

Suppose after n repetitions, where n is very large, an event A occurs k times. Then,

$$P(A) = \frac{k}{n}.$$

Example 2.1. A die is tossed 100 times, and comes up '1' 20 times. If the die is tossed again, what is the probability that it will come up (a) '1', (b) any one of $\{2, 3, 4, 5, 6\}$?

Soln. (a) Based on the results of the experiment,

$$P(\text{getting '1'}) = \frac{20}{100} = 0.2.$$

(b) Since '1' came up 20 times, other numbers $\{2, 3, 4, 5, 6\}$ came up $(100 - 20) = 80$ times. Hence,

$$P(\text{getting any one of } \{2, 3, 4, 5, 6\}) = \frac{80}{100} = 0.8.$$

2.2. Equally Likely Outcomes Approach

Let the sample space S contain n sample points. Suppose each sample point in S is *equally likely to occur*. Then,

$$P(a\ sample\ point\ occurs) = \frac{1}{n}.$$

Let A be an event, $A \subset S$ and there are k sample points in A. Then,

$$P(A) = \frac{k}{n}.$$

Example 2.2. In a fruit basket, there are 5 apples, 3 bananas, 4 oranges and 6 pears. If one fruit is chosen at random, what is the probability that (a) an apple, (b) an orange or a banana, (c) a banana or a pear is chosen?

Soln. (a) Note that each fruit is *equally likely* to be chosen. There are altogether $(5+3+4+6)=18$ fruits.

$$P(\text{an apple is chosen}) = \frac{5}{18}.$$

(b) $P(\text{an orange or a banana is chosen}) = \frac{4+3}{18} = \frac{7}{18}$.

(c) $P(\text{a banana or a pear is chosen}) = \frac{3+6}{18} = \frac{1}{2}$.

Example 2.3. A marble is drawn from a box of 52 marbles. There are 4 red marbles and 12 green marbles in the box. Find the probability p that the marble is (a) a red marble, (b) a green marble.

Soln. (a) There are four red marbles. Hence,

$$p = 4/52 = 1/13.$$

(b) There are 12 green marbles. Hence,

$$p = 12/52 = 3/13.$$

2.3. Finite Sample Space

Consider a finite sample space S with n elements

$$S = \{A_1, A_2, \cdots, A_n\}.$$

Then,

(a) $0 \leq P(A_j) \leq 1, \ j = 1, 2, \cdots, n$

(b) $P(S) = \sum_j P(A_j) = P(A_1) + P(A_2) + \cdots + P(A_n) = 1$

Sometimes the points in a finite sample space S and their assigned probabilities are given in the form of a table as follows:

Outcome	A_1	A_2	$\cdots$	A_n
Probability	$P(A_1)$	$P(A_2)$	$\cdots$	$P(A_n)$

Such a table is called a *probability distribution*.

Summary

1. **Probability is a measure of uncertainty associated with future outcomes of an experiment.**

2. **The *probability of an event* A, $P(A)$, is a numerical measure of the likelihood of the event occurring. $P(A)$ satisfies $0 \leq P(A) \leq 1$. It is equal to the sum of the probabilities of the sample points in A.**

3. **The rule we use in computing probabilities, assuming $equal\ likelihood$ of all sample points, is as follows:**

$$P(A) = \frac{n(A)}{n(S)}$$

where

$n(A) =$ **the no. of elements in the set of the event** A**,**

$n(S) =$ **the no. of elements in the sample space** S**.**

Some probability rules:

(a) (*Mutually exclusive events*) Let A and B be mutually exclusive events, i.e., $A \cap B = \emptyset$. Then,

$$P(A \cup B) = P(A) + P(B).$$

Generalization: If A, B, C are mutually exclusive events, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C).$$

(b) (*General*) **Let** A **and** B **be any 2 events. Then**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

(c) $P(\emptyset) = 0$

(d) (*Complementary event*) **Let** A **be an event. Then,**

$$P(A') = 1 - P(A)$$

or

$$P(A) + P(A') = 1.$$

Example 2.5. The probability that in any week a store will sell 5–10, 11–15, 16–20, and more than 20 computers is 0.2, 0.35, 0.25 and 0.12 respectively. Find the probability that in a given week the store sells at least 11 computers.

Soln. Note that the events of selling 5–10, 11–15, 16–20, and more than 20 computers have no intersection and therefore they are mutually exclusive events. Hence, the probability required is

$$0.35 + 0.25 + 0.12 = 0.72.$$

Example 2.6. In tossing a fair die, what is the probability of getting an even number or a number less than 5?

Soln. Let A be the event '*even number*' and B be the event '*number less than 5*'. Clearly,

$$S = \{1, 2, 3, 4, 5, 6\}, \quad A = \{2, 4, 6\}, \quad B = \{1, 2, 3, 4\}.$$

Method 1: Since $A \cup B = \{1, 2, 3, 4, 6\}$, we have

$$P(A \cup B) = 5/6.$$

Method 2: Use

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Note

$$S = \{1, 2, 3, 4, 5, 6\}, \quad A = \{2, 4, 6\}, \quad B = \{1, 2, 3, 4\}.$$

Since $A \cap B = \{2, 4\}$, we have $P(A \cap B) = 2/6$. Hence,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{4}{6} - \frac{2}{6} = \frac{5}{6}.$$

Example 2.7. Toss three coins and observe the no. of heads. The sample space is $S = \{0, 1, 2, 3\}$. We obtain a probability space by the following assignment

$$P(0) = \frac{1}{8},\ P(1) = \frac{3}{8},\ P(2) = \frac{3}{8},\ P(3) = \frac{1}{8}.$$

Note that the sum of probabilities is 1. Let

A be the event that at least one head appears,

B be the event that all heads or tails appear.

Find $P(A)$, $P(B)$, $P(A \cap B)$ and $P(A \cup B)$. Also find the

probability that not all the 3 tosses come up with the same face.

Soln. Clearly, $S = \{0, 1, 2, 3\}$, $A = \{1, 2, 3\}$ and $B = \{0, 3\}$. Then,

$$P(A) = P(1) + P(2) + P(3) = \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$$

$$P(B) = P(0) + P(3) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$P(A \cap B) = P(3) = \frac{1}{8}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{7}{8} + \frac{1}{4} - \frac{1}{8} = 1$$

or $P(A \cup B) = P(S) = 1$

$P($**not all the 3 tosses come up with the same face**$)$

$$= P(B') = 1 - P(B) = 1 - \frac{1}{4} = \frac{3}{4}$$

3. Independence of Events

Two events are *independent* if the occurrence of one of the events does not change the probability of occurrence of the other event. If the probability is affected, the events are *dependent*.

- **If a coin is flipped twice, the outcome of the 2nd flip does not depend on whether the 1st flip comes up head or tail. The 1st flip and the 2nd flip are independent.**

- If two dice are tossed, the number on the 2nd dice is independent of the number that comes up on the 1st dice. The 1st toss and the 2nd toss are independent.

- Suppose a ball is chosen at random from a bag containing 2 red and 4 blue balls, its color is observed, and then the ball is *replaced* in the bag and the bag is well shaken. Now a 2nd ball is drawn from the bag. What is the probability that the 2nd ball is red?

The answer (2/6) does not depend on the 1st ball drawn, since that ball was replaced, the two draws are independent – the outcome of one does not affect the outcome of the other.

If, however, the 1st ball is *not replaced* before the 2nd ball is drawn, what is the probability that the 2nd ball is red?

Before drawing, the bag contains



(i) If the 1st ball drawn was red, then we have



P(**the 2nd ball is red**) $= 1/5$.

(ii) If the 1st ball drawn was blue, then we have



P(**the 2nd ball is red**) $= 2/5$.

The result of the second draw depends on what happened on the first. We say the two draws are dependent.

To determine whether 2 events are independent/dependent

Two events A and B are said to be *independent* if and only if

$$P(A \text{ and } B) = P(A \cap B) = P(A)P(B).$$

(if and only if means equivalent, notation $\Longleftrightarrow$)

i.e.,

$$A \text{ and } B \text{ are independent } \Longleftrightarrow P(A \cap B) = P(A)P(B).$$

Your Fingerprint represents You

Your Mom is A Woman

A **and** B **are independent** $\iff P(A \cap B) = P(A)P(B)$.

How to <u>check</u> whether $P(A \cap B) = P(A)P(B)$ **??**

Need to calculate 3 probabilities: $P(A),\ P(B),\ P(A \cap B)$

Example *1. Peter and John just graduated from college and started working. The probability that Peter gets promoted a year later is 0.6, while the probability that John gets his promotion a year later is 0.8. Assuming that the career of one has no effect on the career of the other, what is the probability that both of them get promoted a year later?

Soln. Since the career of one does not affect the career of the other, the events are **independent**.

P(Peter and John get promoted a year later)

EXTRA

$= P(\text{Peter gets promoted a year later})$

$\times P(\text{John gets promoted a year later})$

$= (0.6)(0.8) = 0.48.$

Example *2. Let

$A =$ event that a family has children of both sexes,

$B =$ event that a family has at least one boy.

Are A and B independent events if a family has two children?

Soln. We have the sample space $S = \{bb, bg, gb, gg\}$. Each sample point has the same probability.

$$A = \{bg, gb\} \Rightarrow P(A) = \frac{2}{4},$$

$$B = \{bg, gb, bb\} \Rightarrow P(B) = \frac{3}{4},$$

$$A \cap B = \{bg, gb\} \Rightarrow P(A \cap B) = \frac{2}{4}.$$

Since $P(A \cap B) \neq P(A)P(B)$, A and B are dependent events.

Example *3. Toss 2 fair dice simultaneously. Let

A be the event that the number 2 turns up on the first dice,

B be the event that the sum of two dice is 8.

Are the events A and B independent?

Soln. We have the sample space

$$S = \{(i, j) \mid i, j = 1, 2, \cdots, 6\}.$$

Each sample point has the same probability $(= \frac{1}{36})$.

EXTRA

$$A = \{(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)\}$$

$$\Longrightarrow P(A) = \frac{6}{36} = \frac{1}{6}$$

$$B = \{(2,6),(3,5),(4,4),(6,2),(5,3)\}$$

$$\Longrightarrow P(B) = \frac{5}{36}$$

$$A \cap B = \{(2,6)\}$$

$$\Longrightarrow P(A \cap B) = \frac{1}{36}$$

Check: Is $P(A \cap B) = P(A)P(B)$??

Conclusion: A and B are NOT independent.

EXTRA

** **Think** : **Mutually exclusive events** vs **independent events**

Let A and B be 2 events.

Can A and B be $BOTH$ mutually exclusive and independent?

Summary of independent events

1. **Two events A and B are *independent* of one another if the probability of one event does not change whenever the other event occurs.**

2. **A and B are *independent* if and only if ($\Longleftrightarrow$)**

$$P(A \cap B) = P(A)P(B)$$

4. Conditional Probability

Suppose we are told that event B has occurred and we are asked for the probability of event A occurring. This is called the *conditional probability of A given B*, $P(A|B)$.

e.g. $B = \{\text{you study hard}\}$, $A = \{\text{you get good grades}\}$

$P(A|B) = P(\text{you get good grades} \mid \text{you study hard}\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \text{ and } B)}{P(B)}$$

Similarly, the *conditional probability of B given A* is $P(B|A)$,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B \text{ and } A)}{P(A)}.$$

Note: If A and B are *independent events*, then $P(A \cap B) = P(A)P(B)$, and so

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A),$$

i.e., the conditional probability of A given B does not depend on B at all - this justifies the term '*independent*'. Likewise,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)P(B)}{P(A)} = P(B).$$

Example 4.1. In a country club,

75% of members play golf,

65% of members play tennis,

50% of members play both golf and tennis.

A club member is selected at random. (a) If he plays golf, what is the probability that he also plays tennis? (b) If he plays tennis, what is the probability that he also plays golf? (c) What is the probability that he plays golf or tennis?

Soln. Let

$G = \{$a member plays golf$\}$,

$T = \{$a member plays tennis$\}$.

Then,

$$P(G) = 0.75, \quad P(T) = 0.65, \quad P(G \cap T) = 0.5.$$

(a) If he plays golf, what is the probability that he also plays tennis?

$$P(T|G) = \frac{P(T \cap G)}{P(G)} = \frac{0.5}{0.75} = \frac{2}{3}.$$

(b) If he plays tennis, what is the probability that he also plays golf?

$$P(G|T) = \frac{P(G \cap T)}{P(T)} = \frac{0.5}{0.65} = \frac{10}{13}.$$

(c) What is the probability that he plays golf or tennis?

$$P(G \text{ or } T)$$

$$= P(G \cup T) = P(G) + P(T) - P(G \cap T)$$

$$= 0.75 + 0.65 - 0.5 = 0.9.$$

Example 4.2. A roll of 2 fair dice has shown the number 3 on one of the dice. We do not know what the other is. What is the probability that the sum of the 2 numbers is 8?

Soln. Let

$A = \{\text{a 3 turns up}\}$,

$B = \{\text{the sum of the 2 numbers is 8}\}$.

The required probability is $P(B|A)$.

Clearly, the sample space is (see Example 1.6)

$S = \{(i,j) \ : \ i,j = 1,2,\cdots,6\}$ (36 sample points).

$A = \{(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),$
$(1,3),(2,3),(4,3),(5,3),(6,3)\}$ (11 sample points),

$B = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$ (5 sample points),

$A \cap B = \{(3,5),(5,3)\}$ (2 sample points).

Hence,

$$P(A) = \frac{11}{36}, \quad P(B) = \frac{5}{36}, \quad P(A \cap B) = \frac{2}{36}.$$

The required probability is

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{2}{11}.$$

Example *1. Toss a coin three times and note the sequence of heads (h) and tails (t).

(a) What is the probability of getting at least 1 h, given that there is at least 2 t?

(b) It is known that the last toss is a h, what is the probability of getting at most 2 t?

EXTRA

Soln. Clearly, the sample space is

$$S = \{hhh,\ hht,\ hth,\ thh,\ tth,\ tht,\ htt,\ ttt\}$$

(a) What is the probability of getting at least 1 h, given that there is at least 2 t?

$$A = \{\text{getting at least 1 } h\}$$

$$B = \{\text{getting at least 2 } t\}$$

The required probability is $P(A|B)$.

$$A \cap B = \{tth,\ tht,\ htt\}$$

EXTRA

$$P(A) = \frac{7}{8}, \quad P(B) = \frac{4}{8}, \quad P(A \cap B) = \frac{3}{8}, \quad P(A|B) = \frac{3}{4}.$$

(b) It is known that the last toss is a h, what is the probability of getting at most 2 t?

$C = \{\text{the last toss is a } h\}$

$D = \{\text{getting at most 2 } t\}$

The required probability is $P(D|C)$.

$C \cap D = \{hhh,\ hth,\ thh,\ tth\}$

$$P(C) = \frac{4}{8}, \quad P(D) = \frac{7}{8}, \quad P(C \cap D) = \frac{4}{8}, \quad P(D|C) = 1.$$

EXTRA

4.1. Multiplication Rule

By definition of conditional probability, we have

$$\begin{cases} P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A|B)P(B) \\ \\ P(B|A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(A \cap B) = P(B|A)P(A) \end{cases}$$

Hence, $P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$.

This gives us a formula for the probability that events A and B both occur.

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

Recall: A **and** B **are** *independent* **if and only if** ($\Longleftrightarrow$)

$$P(A \cap B) = P(A)P(B)$$

Example 4.3. A quality control procedure for testing bulbs consists of drawing two bulbs at random from each lot of 1000 *without replacement*. If a lot contains 20 defective bulbs, find the probability that both bulbs drawn are (a) defective, (b) non-defective.

Soln. Let

$A = \{$a defective bulb on 1st draw$\}$,

$B = \{$a defective bulb on 2nd draw$\}$.

(a) P(both bulbs drawn are defective)

$$= P(A \cap B) = P(B|A)P(A)$$

Clearly, $P(A) = \frac{20}{1000}$.

The 1st bulb drawn is not replaced, so there are only 999 bulbs for the 2nd draw. If the 1st bulb is defective (A), then there are only 19 defective bulbs left among the 999 remaining bulbs in the lot. Thus, $P(B|A) = \frac{19}{999}$.

Hence, P(both bulbs drawn are defective)

$$= P(A \cap B) = P(B|A)P(A) = \frac{19}{999}\ \frac{20}{1000} = 0.00038.$$

(b) P(both bulbs drawn are non-defective)

$$= P(A' \cap B') = P(B'|A')P(A')$$

Clearly, $P(A') = 1 - P(A) = 1 - \frac{20}{1000} = \frac{980}{1000}$.

If the 1st bulb is non-defective (A')**, then there are 20 defective bulbs among the 999 remaining bulbs in the lot. Thus,** $P(B'|A') = \frac{999-20}{999} = \frac{979}{999}$.

Hence, P(both bulbs drawn are defective)

$$= P(A' \cap B') = P(B'|A')P(A') = \frac{979}{999}\ \frac{980}{1000} = 0.96.$$

General multiplication rule

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

$$P(A \cap B \cap C \cap D) = P(A)P(B|A)P(C|A \cap B)P(D|A \cap B \cap C)$$

EXTRA

Example *2. Suppose that 5 good and 2 defective fuses have been mixed up. To find the defective ones, we test them one-by-one, at random and without replacement. What is the probability that we find both of the defective fuses in exactly 3 tests.

Soln. For $i = 1, 2, 3$, let

$D_i = \{$the i-th fuse tested is defective$\}$

$G_i = \{$the i-th fuse tested is good$\}$

Prob required $= P(G_1 \cap D_2 \cap D_3) + P(D_1 \cap G_2 \cap D_3)$

EXTRA

Noting

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

Prob required

$$\begin{aligned}
&= P(G_1 \cap D_2 \cap D_3) + P(D_1 \cap G_2 \cap D_3) \\
&= P(G_1)P(D_2|G_1)P(D_3|G_1 \cap D_2) \\
&\qquad + P(D_1)P(G_2|D_1)P(D_3|D_1 \cap G_2) \\
&= \frac{5}{7}\frac{2}{6}\frac{1}{5} + \frac{2}{7}\frac{5}{6}\frac{1}{5}
\end{aligned}$$

EXTRA

Summary of conditional probability

1. The *conditional probability of B given A* is $P(B|A)$,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B \text{ and } A)}{P(A)}.$$

If A and B are independent events, then

$$P(B|A) = P(B),\ P(A|B) = P(A).$$

2. Multiplication rule

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

If A and B are independent events, then the multiplication rule reduces to

$$P(A \cap B) = P(A)P(B).$$

5. Partitions, Total Probability, Bayes Theorem

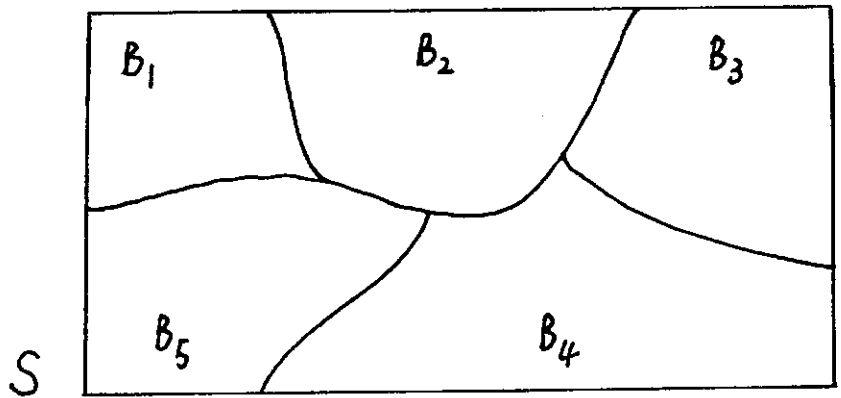
- If

  $B_1, B_2, \cdots, B_k$ are disjoint subsets of S,

  and

  $B_1 \cup B_2 \cup \cdots \cup B_k = S$,

  then $\{B_1, B_2, \cdots, B_k\}$ is a *partition* of S.


B_1
B_2
B_3
B_5
B_4
S

- **Let A be an event of S. We can write**

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_k).$$

Since $(A \cap B_j)$'s are mutually exclusive, we have

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_k).$$

Using $P(A \cap B_j) = P(A|B_j)P(B_j)$, it follows that

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k).$$

- **Total Probability Theorem**

  Let $\{B_1, B_2, \cdots, B_k\}$ be a partition of S, and A be an event of S. Then, the *total probability* of A is

$$\begin{aligned} & P(A) \\ &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k) \\ &= \sum_{j=1}^{k} P(A|B_j)P(B_j). \end{aligned}$$

- **Bayes Theorem**

  Let $\{B_1, B_2, \cdots, B_k\}$ be a partition of S, and A be an event of S. Then, for $r = 1, 2, \cdots, k$,

$$P(B_r|A) = \frac{P(A|B_r)P(B_r)}{\sum_{j=1}^{k} P(A|B_j)P(B_j)}.$$

  Proof. By the definition of conditional probability, we have

$$P(B_r|A) = \frac{P(A \cap B_r)}{P(A)} = \frac{P(A|B_r)P(B_r)}{P(A)}.$$

Now, apply Total Probability Theorem to the denominator. This gives

$$P(B_r|A) = \frac{P(A|B_r)P(B_r)}{\sum_{j=1}^{k} P(A|B_j)P(B_j)}. \qquad \text{(end of proof)}$$

Example 5.1. A marble will be picked from two boxes containing some marbles. The *square* box contains 3 red and 2 blue marbles while the *round* box contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up heads, a marble is picked from the *square* box; if it turns up tails,

a marble is picked from the $round$ box. Find the probability that a red marble is chosen.

Soln. Let

$A = \{$a red marble is chosen$\}$

$B_1 = \{$a marble is picked from the $square$ box$\}$

$B_2 = \{$a marble is picked from the $round$ box$\}$

Note that $\{B_1, B_2\}$ forms a partition of S (What is S?)

By Total Probability Theorem,

$$\begin{aligned} P(A) &= \sum_{j=1}^{k} P(A|B_j)P(B_j) \\ &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) \end{aligned}$$

Clearly,

$$P(B_1) = 0.5 \qquad P(B_2) = 0.5$$

$$P(A|B_1) = \frac{3}{3+2} \qquad P(A|B_2) = \frac{2}{2+8}$$

Therefore, $P(A) = 0.6 \times 0.5 + 0.2 \times 0.5 = 0.4.$

Example 5.2. In Example 5.1, suppose we do not know whether the coin turns up heads or tails, but we are told that a red marble has been chosen.

(a) What is the probability that the red marble is from the *square* box (i.e. the coin turns up heads)?

(b) What is the probability that the red marble is from the *round* box (i.e. the coin turns up tails)?

Soln. (a) Recall

$A = \{$a red marble is chosen$\}$

$B_1 = \{$a marble is picked from the $square$ box$\}$

$B_2 = \{$a marble is picked from the $round$ box$\}$

$\{B_1, B_2\}$ is a partition, so Bayes Theorem can be used.

$$P(B_r|A) = \frac{P(A|B_r)P(B_r)}{\sum_{j=1}^{k} P(A|B_j)P(B_j)}$$

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A|B_1)P(B_1)+P(A|B_2)P(B_2)}$$

$$= \frac{P(A|B_1)P(B_1)}{P(A)} = \frac{(0.6)(0.5)}{0.4} = 0.75$$

(b)

$$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A|B_1)P(B_1)+P(A|B_2)P(B_2)}$$

$$= \frac{P(A|B_2)P(B_2)}{P(A)}$$

$$= \frac{(0.2)(0.5)}{0.4} = 0.25$$

Note $P(B_1|A) + P(B_2|A) = 1$. This is clear since

$$P(B_1|A) + P(B_2|A) = \frac{P(A|B_1)P(B_1)}{P(A|B_1)P(B_1)+P(A|B_2)P(B_2)}$$

$$+ \frac{P(A|B_2)P(B_2)}{P(A|B_1)P(B_1)+P(A|B_2)P(B_2)} = 1.$$

In general, if $\{B_1, B_2, \cdots, B_k\}$ is a partition, then

$$\sum_{j=1}^{k} P(B_j|A) = 1.$$

Example 5.3. Three alarms are installed at a house to prevent break-in. Let

$$B_j = \{\text{alarm } j \text{ is triggered}\}, \qquad j = 1, 2, 3$$

$$A = \{\text{house is broken in}\}.$$

The chance of triggering each alarm is

$$P(B_1) = 0.1, \qquad P(B_2) = 0.2, \qquad P(B_3) = 0.05.$$

The probability that the house is broken in if a particular alarm is triggered is

$$P(A|B_1) = 0.1, \quad P(A|B_2) = 0.2, \quad P(A|B_3) = 0.3.$$

If a thief breaks into the house, which alarm is most likely to be triggered?

Soln. Use **Bayes Theorem**.

$$\sum_{j=1}^{3} P(A|B_j)P(B_j) = (0.1)(0.1) + (0.2)(0.2) + (0.3)(0.05)$$

$$= 0.065$$

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{\sum_{j=1}^{3} P(A|B_j)P(B_j)} = \frac{(0.1)(0.1)}{0.065} = 0.15$$

$$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{\sum_{j=1}^{3} P(A|B_j)P(B_j)} = \frac{(0.2)(0.2)}{0.065} = 0.62$$

$$P(B_3|A) = \frac{P(A|B_3)P(B_3)}{\sum_{j=1}^{3} P(A|B_j)P(B_j)} = \frac{(0.3)(0.05)}{0.065} = 0.23$$

Since $P(B_2|A)$ is the highest, alarm number 2 is most likely to be triggered.

** **Think 2** : **Refer to Example 5.3 (Bayes Theorem)**

1. **Does $\{B_1, B_2, B_3\}$ form a partition of S? Give reasons.**

2. **Why is it that we can apply Bayes Theorem here?**

Example *1. A box contains books in 3 different languages, namely Chinese, English, French, and some of the books are hard-cover books. The probability that a Chinese book has hard cover is 0.8, with the corresponding probabilities for English and French books being 0.5 and 0.3, respectively. About 30% of the books in the box are Chinese, 40% are English, and 30% are French.

(a) Find the probability that a book randomly selected from the box has hard cover?

EXTRA

(b) A book was picked from the box and it was a hard-cover book. What is the probability that the book was a French book?

<u>Soln.</u> (a) Let

$A = \{$a hard-cover book is chosen$\}$

$B_1 = \{$a Chinese book is chosen$\}$

$B_2 = \{$an English book is chosen$\}$

$B_3 = \{$a French book is chosen$\}$

EXTRA

From the question, we get

$$P(A|B_1) = 0.8, \quad P(A|B_2) = 0.5, \quad P(A|B_3) = 0.3,$$

$$P(B_1) = 0.3, \quad P(B_2) = 0.4, \quad P(B_3) = 0.3.$$

To compute $P(A)$, we use total probability of A, to obtain

$$\begin{aligned} P(A) &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) \\ &= 0.8 \times 0.3 + 0.5 \times 0.4 + 0.3 \times 0.3 \\ &= 0.53. \end{aligned}$$

EXTRA

(b) From Bayes Theorem, we obtain

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{P(A)}, \quad j = 1, 2, 3.$$

Thus, we have

$$P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = \frac{0.3 \times 0.3}{0.53} = 0.170.$$

EXTRA

Example *2. In Example *1, suppose another box also contains books in 3 different languages, namely Chinese (50%), English (30%), French (20%), and some of the books are hard-cover books. If we pick a book randomly from one of the boxes, what is the probability that we get a Chinese book?

Soln. Corresponding to B_j in Example *1, let E_j be the event that a Chinese/English/French book is chosen from the second box, $j = 1, 2, 3$. From the question, we get

$$P(E_1) = 0.5, \quad P(E_2) = 0.3, \quad P(E_3) = 0.2.$$

EXTRA

Let

$F_1 =$ {a book is chosen from the 1st box}

$F_2 =$ {a book is chosen from the 2nd box}

$C =$ {a Chinese book is chosen}

$P(F_1) = 0.5, \quad P(F_2) = 0.5.$

By Total Probability Theorem,

$$\begin{aligned} P(C) &= P(C|F_1)P(F_1) + P(C|F_2)P(F_2) \\ &= 0.3 \times 0.5 + 0.5 \times 0.5 = 0.4. \end{aligned}$$

EXTRA

Summary

1. **Total Probability Theorem**: Let $\{B_1, B_2, \cdots, B_k\}$ be a partition of S, and A be an event of S. Then,

$$P(A) = \sum_{j=1}^{k} P(A|B_j)P(B_j).$$

2. **Bayes Theorem**: Let $\{B_1, B_2, \cdots, B_k\}$ be a partition of S, and A be an event of S. Then, for $r = 1, 2, \cdots, k$,

$$P(B_r|A) = \frac{P(A|B_r)P(B_r)}{\sum_{j=1}^{k} P(A|B_j)P(B_j)}.$$

7. Random Variables

The topic of random variables (r.v.) is fundamental to probability and statistics.

The outcome of an experiment need not be a number. However, we can assign a numerical value to each of the outcomes of the experiment. Since the outcomes are random, the numerical values obtained will also be random.

$e.g.$ we are interested in the gender of the next person who walks into the LT, the 2 outcomes are M, F. We can assign 1 to outcome M, and 2 to outcome F.

$e.g.$ we are conducting a survey of the no. of pens a NTU student carries to school. The sample space $S = \{0, 1, 2, \cdots\}$. We can assign (naturally) the number k to outcome 'k'.

- A *random variable*, X, is a function that assigns a real number, $X(\xi)$, to each outcome ξ in the sample space S of an experiment.

- A r.v. is one which can take a number of different values, but exactly which value the variable does take is not known until some 'experiment' is performed.

- We use a capital letter such as X to denote a r.v., and its lower-case counterpart x to denote a particular value.

- **Examples of random variables:**

(a) Throwing a pair of dice. $X =$ sums.

(b) Questioning a sample of 200 students about an issue. $X =$ the no. of students in the sample taking a positive stand.

(c) Measuring the height of a randomly selected student. $X =$ the height observed.

(d) Finding the average daily expenditure of a sample of 20 students. $X =$ the mean of the sample.

Example 7.1. Roll a dice. There are six possible outcomes whose numeric values are 1, 2, 3, 4, 5 and 6. If we let X be a r.v. which measures the no. of spots uppermost on the dice after it has been rolled, then X can take the values 1, 2, 3, 4, 5 and 6.

7.1. Discrete and Continuous r.v.'s

- If the values of a r.v. can be counted, the r.v. is called *discrete*.

- If the values of the r.v. cannot be counted, the r.v. is called *continuous*.

- **Some examples:**

(a) Throwing a pair of dice. $X =$ sums. X is discrete.

(b) Questioning a sample of 200 students about an issue.

$X =$ the no. of students in the sample taking a positive stand. X is discrete.

(c) Measuring the height of a randomly selected student.

$X =$ the height observed. X is continuous.

(d) Finding the average daily expenditure of a sample of 20 students.

$X =$ the mean of the sample. X is continuous.

Summary

1. A *random variable* (r.v.) is an uncertain quantity whose value depends on chance. It is a rule for assigning real numbers to the outcomes of an experiment.

2. A r.v. is called *discrete* if all of its possible values are countable.

3. A *continuous* r.v. may take on any value in an interval of numbers (i.e., its possible values are uncountably infinite).

7.2. Probability Distribution

A r.v. has a probability law: A rule that assigns probabilities to the different values of the r.v. The law is called the *probability distribution* of the r.v.

X is a *discrete r.v.*

$P(k) \equiv P(X = k)$ is known as the *probability mass function* (pmf) of the discrete r.v. X. $P(k)$ associates a probability with each numerical value k that X can assume. It must satisfy the following two conditions:

1. $0 \leq P(k) \leq 1$ for all possible values of k,

2. $\sum_{k} P(k) = 1$.

Example 7.2. Let the r.v. X be the no. of heads obtained when two coins are tossed. The possible values for X are 0, 1, 2. The sample space

$$S = \{hh,\ ht,\ th,\ tt\}.$$

Clearly,

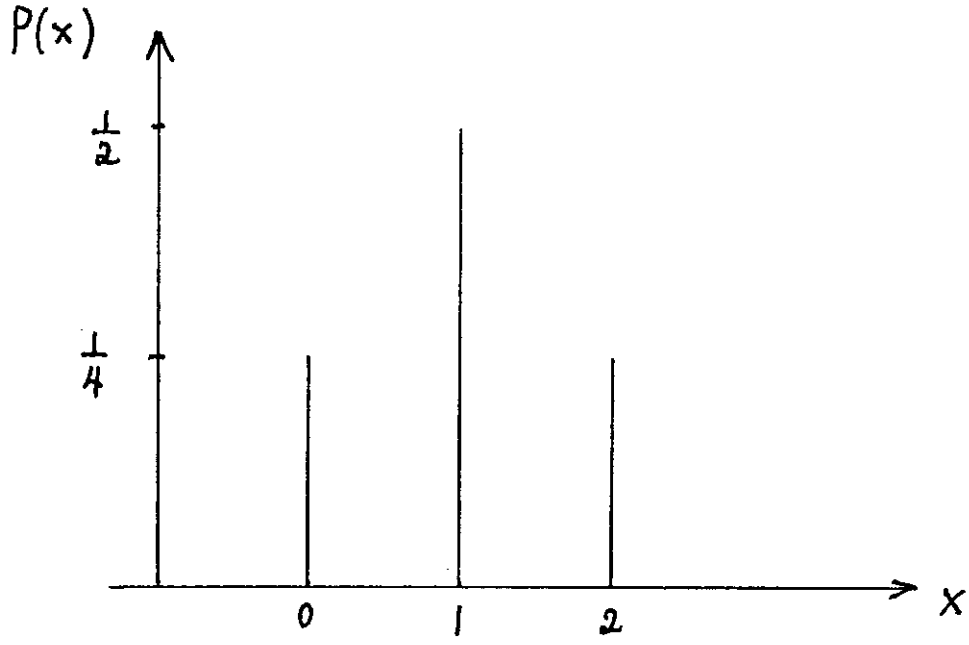
$$P(X = 0) = P(0) = P(\{tt\}) = 1/4,$$
$$P(X = 1) = P(1) = P(\{ht, th\}) = 1/2,$$
$$P(X = 2) = P(2) = P(\{hh\}) = 1/4.$$

Check: $P(0) + P(1) + P(2) = 1.$

The *probability distribution table* for the discrete r.v. X is shown below:

x	0	1	2
P(x)	1/4	1/2	1/4

The *probability distribution* in graphic form is as shown:


P(x)
1/2
1/4
0
1
2
X

X is a *continuous r.v.*

For continuous r.v.'s, we use areas instead of vertical line segments to represent probabilities.

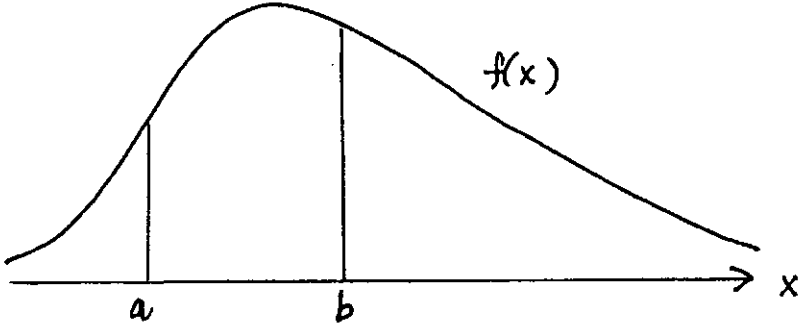
The *probability density function* (pdf) of a continuous r.v. X is the function that associates probabilities with intervals over which X is defined. The function, denoted by f, has the following properties:

1. $f(x) \geq 0$ for all x

2. $P(a < X < b) = \int_a^b f(x)dx$

   The probability $P(a < X < b)$ equals the area under the curve of the probability density function f between a and b.

$$P(a < X < b) = \int_a^b f(x)dx$$



3. $\int_{-\infty}^{\infty} f(x)dx = 1$

4. $P(a < X < b) = \int_{a}^{b} f(x)dx$

$$P(a < X < b) = P(a \leq X \leq b) = P(a \leq X < b)$$
$$= P(a < X \leq b)$$

$$P(X = a) = 0$$

These are not true, though, when X is *discrete*.

Example 7.3. Suppose the values of the r.v. X are all real numbers between 0 and 1. The probability density function of X is f, defined by

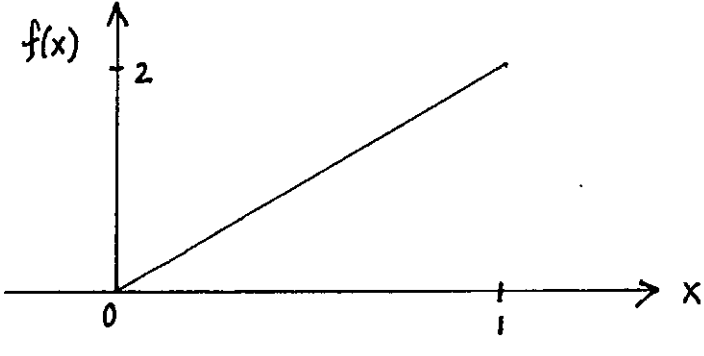
$$f(x) = \begin{cases} 2x, & if\ 0 \leq x \leq 1 \\ 0, & elsewhere. \end{cases}$$

A graph of f is shown below:

Check: $\int_{-\infty}^{\infty} f(x)dx = \int_{0}^{1} f(x)dx = \int_{0}^{1} 2xdx = 1.$

$$f(x) = \begin{cases} 2x, & if\ 0 \leq x \leq 1 \\ 0, & elsewhere. \end{cases}$$

A graph of f is shown below:



Example *1. Let $f(x)$ be the probability density function of a random variable X given by

$$f(x) = \begin{cases} c\left(x^{-1/2}\right), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

where c is a constant.

(a) Find c.

(b) Find the probability $P\left(X \leq \frac{1}{4}\right)$.

Soln. (a) We have

$$\int_{-\infty}^{\infty} f(x)dx = 1 \;\Rightarrow\; \int_{0}^{1} c\left(x^{-1/2}\right)dx = 1$$

$$\Rightarrow\; c\left[\frac{x^{1/2}}{1/2}\right]_{0}^{1} = 1 \;\Rightarrow\; 2c = 1 \;\Rightarrow\; c = \frac{1}{2}$$

(b) $$P\left(X \le \frac{1}{4}\right) = \int_{0}^{1/4} f(x)dx = \int_{0}^{1/4} c\left(x^{-1/2}\right)dx$$

$$= c\left[\frac{x^{1/2}}{1/2}\right]_{0}^{1/4} = c = \frac{1}{2}$$

Example *2. Let $f(x)$ be the probability density function of a random variable X given by

$$f(x) = \begin{cases} \alpha x + 1, & 0 \leq x \leq \beta \\ 0, & \text{otherwise} \end{cases}$$

where α and β are positive constants.

(a) Find the relation between two constants α and β. Hence, compute the value of β when $\alpha = 4$.

(b) Find $P\left(X \geq \frac{1}{2}\beta\right)$ and $P\left(X \leq \frac{3}{4}\beta \middle| X \geq \frac{1}{2}\beta\right)$ when $\alpha = 4$.

EXTRA

Soln. (a) From the property $\int_{-\infty}^{\infty} f(x)dx = 1$, it follows that

$$\int_0^d (\alpha x + 1)dx = 1 \Rightarrow \left[\frac{\alpha x^2}{2} + x\right]_0^{\beta} = 1$$

$$\Rightarrow \alpha\beta^2 + 2\beta - 2 = 0.$$

When $\alpha = 4$, the above relation becomes

$$2\beta^2 + \beta - 1 = 0 \Rightarrow (2\beta - 1)(\beta + 1) = 0 \Rightarrow \beta = \frac{1}{2} \text{ or } -1.$$

Notice that $0 \leq x \leq \beta$, so $\beta \geq 0$. Therefore, we should have $\beta = \frac{1}{2}$ when $\alpha = 4$.

EXTRA

(b) $P\left(X \geq \frac{1}{2}\beta\right) = \int_{\beta/2}^{\beta} f(x)dx = \int_{1/4}^{1/2} (4x+1)dx$

$$= \left[2x^2 + x\right]_{1/4}^{1/2} = \frac{5}{8}$$

(c) $P\left(X \leq \frac{3}{4}\beta \middle| X \geq \frac{1}{2}\beta\right) = \dfrac{P\left(\frac{1}{2}\beta \leq X \leq \frac{3}{4}\beta\right)}{P\left(X \geq \frac{1}{2}\beta\right)}.$

Clearly,

$$P\left(\tfrac{1}{2}\beta \leq X \leq \tfrac{3}{4}\beta\right) = \int_{\frac{1}{2}\beta}^{\frac{3}{4}\beta} f(x)dx = \int_{\frac{1}{4}}^{\frac{3}{8}} (4x+1)dx = \frac{9}{32}.$$

EXTRA

Thus, together with the answer of (b) we get

$$P\left(X \leq \tfrac{3}{4}\beta \,\middle|\, X \geq \tfrac{1}{2}\beta\right) = \frac{P\left(\frac{1}{2}\beta \leq X \leq \frac{3}{4}\beta\right)}{P\left(X \geq \frac{1}{2}\beta\right)} = \frac{9/32}{5/8} = \frac{9}{20}.$$

EXTRA

Example *3. The probability mass function of a random variable X is shown in the following table

x	1	2	3	4
$P(x)$	1/8	1/4	c	1/2

where c is a constant.

(a) Find c.

(b) Let A and B be the events that $X = 2$ and $X \geq 3$, respectively. Find $P(A)$, $P(B)$, $P(A \cap B)$. Are events A and B independent?

Soln. (a) From the property $\sum_x P(x) = 1$, it follows that

$$P(1) + P(2) + P(3) + P(4) = 1 \Rightarrow \frac{1}{8} + \frac{1}{4} + c + \frac{1}{2} = 1$$
$$\Rightarrow c = \frac{1}{8}.$$

(b) $P(A) = P(X = 2) = \frac{1}{4}$

$$P(B) = P(X \geq 3) = P(X = 3) + P(X = 4) = \frac{1}{8} + \frac{1}{2} = \frac{3}{8}.$$

$$P(A \cap B) = P(X = 2 \text{ and } X \geq 3) = P(\emptyset) = 0.$$

Since $P(A \cap B) \neq P(A) \cdot P(B)$, events A and B are not independent, i.e., they are dependent.

Summary of pmf, pdf

1. A *random variable* (r.v.) is an uncertain quantity whose value depends on chance. It is a rule for assigning real numbers to the outcomes of an experiment.

2. A r.v. is called *discrete* if all of its possible values are countable.

3. A *continuous* r.v. may take on any value in an interval of numbers (i.e., its possible values are uncountably infinite).

4. **A r.v. has a probability law: A rule that assigns probabilities to the different values of the r.v. The law is called the *probability distribution* of the r.v.**

5. **The *probability mass function* (pmf) of a discrete r.v. X is the function that associates a probability with each numerical value that X can assume. We write $P(k) \equiv P(X = k)$, where k's are the values that X can assume. It must satisfy the following two conditions:**

   (1) $0 \leq P(k) \leq 1$ for all possible values of k

(2) $\sum_{k} P(k) = 1$

6. The *probability density function* (pdf) of a continuous r.v. X is the function that associates probabilities with intervals over which X is defined. The function, denoted by f, has the following properties:

   (1) $f(x) \geq 0$ for all x

   (2) $P(a < X < b) = P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$

(3) $\int_{-\infty}^{\infty} f(x)dx = 1$

Also, $P(X = a) = 0$**, i.e., the probability of any single value is zero.**

7.3. Distribution Functions

Let X be a r.v. (discrete or continuous). The *distribution function* F of X is defined by

$$F(x) = P(X \leq x).$$

- If X is a *discrete r.v.*, then

$$F(x) = \sum_{k \leq x} P(X = k) \qquad \text{(sum over those } k \leq x\text{)}.$$

- If X is a *continuous r.v.*, then

$$\begin{aligned} F(x) &= P(X \leq x) = \int_{-\infty}^{x} f(y)dy, \\ F'(x) &= f(x), \end{aligned}$$

$$\begin{aligned} P(a < X < b) &= \int_{a}^{b} f(x)dx \\ &= \int_{-\infty}^{b} f(x)dx - \int_{-\infty}^{a} f(x)dx \\ &= F(b) - F(a). \end{aligned}$$

Example 7.4. Let X be a continuous r.v. having the probability density function

$$f(x) = \begin{cases} 0.75(1-x^2), & if \ -1 \le x \le 1 \\ 0, & otherwise. \end{cases}$$

(a) Find the distribution function of X.

For $x < -1$,

$$F(x) = P(X \le x) = 0.$$

For $-1 \leq x < 1$,

$$F(x) = \int_{-\infty}^{x} f(y)dy = \int_{-1}^{x} 0.75(1-y^2)dy$$

$$= \ 0.5 + 0.75x - 0.25x^3.$$

For $x \geqslant 1$,

$$F(x) = \int_{-\infty}^{x} f(y)dy = \int_{-1}^{1} f(y)dy = 1.$$

Hence,

$$F(x) = \begin{cases} 0, & x < -1 \\ 0.5 + 0.75x - 0.25x^3, & -1 \leq x < 1 \\ 1, & x \geqslant 1. \end{cases}$$

(b) Find $P(-0.5 \leq X \leq 0.5)$.

$$P(-0.5 \leq X \leq 0.5) = \int_{-0.5}^{0.5} f(y)dy = 0.6875$$

or $P(-0.5 \leq X \leq 0.5) = F(0.5) - F(-0.5) = 0.6875.$

(c) Find $P(-0.25 \leq X \leq 2)$.

$$P(-0.25 \leq X \leq 2) = \int_{-0.25}^{1} f(y)dy = 0.6836 \quad \text{or}$$

$$P(-0.25 \leq X \leq 2) = F(2) - F(-0.25) = 1 - 0.3164 = 0.6836.$$

(d) Find x such that $P(X \leq x) = 0.95$.

$$P(X \leq x) = F(x) = 0.95 \Rightarrow 0.5 + 0.75x - 0.25x^3 = 0.95$$

Solving numerically, x is approximately 0.73.

Example *4. The probability mass function of a random variable X is shown in the following table

x	1	2	3	4
$P(x)$	1/8	1/4	1/8	1/2

Find the distribution function $F(x)$, and sketch the graph of $F(x)$.

Soln. Use the definition of distribution function of discrete random variable, $F(x) = P(X \leq x) = \sum_{x_j \leq x} P(X = x_j)$.

For $x < 1$, $F(x) = 0$.

For $1 \leq x < 2$,

$$F(x) = P(X \leq x) = P(X = 1) = \frac{1}{8}.$$

For $2 \leq x < 3$,

$$\begin{aligned} F(x) = P(X \leq x) &= P(X = 1) + P(X = 2) \\ &= \frac{1}{8} + \frac{1}{4} = \frac{3}{8}. \end{aligned}$$

For $3 \leq x < 4$,

$$\begin{aligned} F(x) = P(X \leq x) &= P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{1}{8} + \frac{1}{4} + \frac{1}{8} = \frac{1}{2}. \end{aligned}$$

For $x \geq 4$,

$$\begin{aligned} F(x) = P(X \leq x) &= \sum_{k=1}^{4} P(X = k) \\ &= \frac{1}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{2} = 1. \end{aligned}$$

Hence, $F(x)$ is given as

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{8}, & 1 \leq x < 2 \\ \frac{3}{8}, & 2 \leq x < 3 \\ \frac{1}{2}, & 3 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Summary of $F(x)$

1. The *distribution function* of a r.v. X (discrete or continuous) is the function

$$F(x) = P(X \leq x)$$

$$= \begin{cases} \displaystyle\int_{-\infty}^{x} f(y)dy, & X \text{ is continuous} \\ \displaystyle\sum_{k=-\infty}^{x} P(X = k), & X \text{ is discrete} \end{cases}$$

2. For a continuous r.v. X,

$$P(a < X < b) = \int_a^b f(x)dx = F(b) - F(a)$$

7.4. Expected Value

- The *expected value* (or mean, average value) of a r.v. X, denoted by μ_X or $E(X)$, is defined by

$$E(X) = \begin{cases} \sum_x xP(x), & X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} xf(x)dx, & X \text{ is continuous.} \end{cases}$$

- $E(X)$ measures where the probability distribution is *centered*.

- **$E(X)$ is the *mean* of the values of the r.v. X over many repeated observations, i.e., $E(X)$ is the *long-run average* of the r.v. X.**

- **Useful properties:**

  Let c be a constant and let X and Y be r.v.'s.

1. $E(c) = c$
2. $E(cX) = cE(X)$
3. $E(X + Y) = E(X) + E(Y)$

From 1–3, it follows that

$$E(aX + bY) = aE(X) + bE(Y)$$

where a, b are constants.

- If X is a r.v. and $g(X)$ is a continuous function of X, then the *expected value* of $g(X)$ is

$$E[g(X)] = \begin{cases} \sum_x g(x)P(x), & X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} g(x)f(x)dx, & X \text{ is continuous.} \end{cases}$$

  When $g(X) = X$, the above reduces to the formula for $E(X)$.

The expected value of a r.v. describes where the probability distribution is centered. By itself, however, the mean does not give adequate information about the shape of the distribution. We need to characterize the variability in the distribution.

e.g. test scores of class A: 47, 48, 49, 51, 52, 53

---

e.g. test scores of class B: 0, 1, 2, 98, 99, 100

7.5. Variance and Standard Deviation

- The *variance* of a r.v. X, denoted by σ_X^2 or $Var(X)$, is the expected value of $(X - \mu_X)^2$, i.e.,

$$Var(X) = E[(X - E(X))^2].$$

Thus,

$$Var(X) = E[(X - E(X))^2]$$

$$= \begin{cases} \sum_x (x - \mu_X)^2 P(x), & X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx, & X \text{ is continuous.} \end{cases}$$

In practice, it is useful to note that

$$Var(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2.$$

To show that $E[(X - E(X))^2] = E(X^2) - [E(X)]^2$

$$
\begin{aligned}
E[(X - E(X))^2] &= E[X^2 - 2XE(X) + (E(X))^2] \\
&= E[X^2 - 2X\mu_X + \mu_X^2] \\
&= E(X^2) - 2\mu_X E(X) + \mu_X^2 \\
&= E(X^2) - 2E(X)E(X) + [E(X)]^2 \\
&= E(X^2) - [E(X)]^2
\end{aligned}
$$

- The *standard deviation* of a r.v. X, denoted by σ_X, is the positive square root of $Var(X)$, i.e.,

$$\sigma_X = \sqrt{Var(X)}.$$

- $Var(X)$ measures the amount of *variability* in the distribution of X, i.e., the spread of a r.v. about its mean. It gives us an idea of the variation or uncertainty associated with the r.v: the larger the variance, the further away from the mean are possible values of the r.v.

- **Useful properties:**

  Let c be a constant and let X and Y be r.v.'s.

  1. $Var(c) = 0$
  2. $Var(cX) = c^2\ Var(X)$
  3. $Var(X + Y) \neq Var(X) + Var(Y)$

Example 7.5. **Let the r.v.'s X and Y be related by $Y = aX + b$ where a, b are constants. Show that**

$$E(Y) = aE(X) + b \quad \text{and} \quad Var(Y) = a^2\ Var(X).$$

Proof. $E(Y) = E(aX+b) = E(aX)+E(b) = aE(X)+b$

$$\begin{aligned} Var(Y) &= Var(aX+b) \\ &= E[(aX+b)^2] - [E(aX+b)]^2 \\ &= E(a^2X^2+2abX+b^2) - [aE(X)+b]^2 \\ &= a^2E(X^2)+2abE(X)+b^2 \\ &\qquad - [a^2(E(X))^2+2abE(X)+b^2] \\ &= a^2[E(X^2)-(E(X))^2] = a^2\ Var(X) \end{aligned}$$

EXTRA

a, b are constants.

$$E(aX+b) = aE(X) + b$$

$$Var(aX+b) = a^2 Var(X)$$

e.g. $E(3X+2) = 3E(X) + 2$

$Var(3X+2) = 3^2 Var(X) = 9Var(X)$

$E(4-5X) = 4 - 5E(X)$

$Var(4-5X) = (-5)^2 Var(X) = 25Var(X)$

Example 7.6. Consider a discrete r.v. X whose probability mass function is given by

$$P(x) = \begin{cases} 1/3, & x = -1, 0, 1 \\ 0, & otherwise. \end{cases}$$

Find the mean and variance of (a) X, (b) $2X - 1$.

Soln. (a) The mean of X is

$$E(X) = \sum_x xP(x) = (-1)\frac{1}{3} + (0)\frac{1}{3} + (1)\frac{1}{3} = 0.$$

Next,

$$E(X^2) = \sum_x x^2 P(x) = (-1)^2 \, \frac{1}{3} + (0)^2 \, \frac{1}{3} + (1)^2 \, \frac{1}{3} = \frac{2}{3}.$$

Thus, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{2}{3} - 0 = \frac{2}{3}.$$

(b) The mean of $(2X - 1)$ is

$$E(2X - 1) = 2E(X) - 1 = 2(0) - 1 = -1.$$

The variance of $(2X-1)$ is

$$Var(2X-1) = 2^2\ Var(X) = (4)\frac{2}{3} = \frac{8}{3}.$$

Example 7.7. Let X be a continuous r.v. whose probability density function is given by

$$f(x) = \begin{cases} cx, & 0 < x < 1 \\ 0, & otherwise \end{cases}$$

where c is a constant. Find (a) c, (b) the mean and variance

of X and $\sqrt{X}$.

<u>Soln.</u> (a) We have $f(x) = \begin{cases} cx, & 0 < x < 1 \\ 0, & otherwise. \end{cases}$

Thus,

$$\int_{-\infty}^{\infty} f(x)dx = 1 \;\Rightarrow\; \int_{0}^{1} cxdx = 1$$

$$\Rightarrow\; c\left[\frac{x^2}{2}\right]_0^1 = 1 \;\Rightarrow\; \frac{c}{2} = 1 \;\Rightarrow\; c = 2.$$

(b) The mean of X is

$$E(X) = \int_0^1 xf(x)dx = \int_0^1 x(2x)dx = \frac{2}{3}.$$

Next,

$$E(X^2) = \int_0^1 x^2 f(x)dx = \int_0^1 x^2(2x)dx = \frac{1}{2}.$$

Hence, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

The mean of $\sqrt{X}$ is

$$E(\sqrt{X}) = \int_0^1 \sqrt{x} f(x) dx = \int_0^1 \sqrt{x}(2x) dx = \frac{4}{5}.$$

Next,

$$E(\sqrt{X}^{\,2}) = E(X) = \frac{2}{3}.$$

Hence, the variance of $\sqrt{X}$ is

$$Var(\sqrt{X}) = E(\sqrt{X}^{\,2}) - [E(\sqrt{X})]^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75}.$$

Example *1. The probability mass function of a random variable X is shown in the following table

x	1	2	3	4
$P(x)$	1/8	1/4	1/8	1/2

Find $E\left(X^2\right)$ and $Var\left(X^2\right)$. Hence, compute $E\left(5-3X^2\right)$ and $Var\left(6-2X^2\right)$.

Soln. Using the formula $E\left(g(X)\right)=\sum_x g(x)P(x)$ gives

$$E\left(X^2\right)=\sum_x x^2P(x)=(1^2)\tfrac{1}{8}+(2^2)\tfrac{1}{4}+(3^2)\tfrac{1}{8}+(4^2)\tfrac{1}{2}=10.25$$

EXTRA

$$E\left(X^4\right) = \sum_x x^4 P(x) = (1^4)\frac{1}{8} + (2^4)\frac{1}{4} + (3^4)\frac{1}{8} + (4^4)\frac{1}{2} = 142.25$$

Using $Var(X) = E(X^2) - [E(X)]^2$, we get

$$Var(X^2) = E(X^4) - [E(X^2)]^2 = 142.25 - 10.25^2 = 37.188$$

Noting $E(aX + b) = aE(X) + b$ and $Var(aX + b) = a^2 Var(X)$, we get

$$E(5 - 3X^2) = 5 - 3E(X^2) = 5 - 3(10.25)$$

EXTRA

$$Var(6-2X^2) = (-2)^2 Var(X^2) = 4Var(X^2) = 4(37.188)$$

EXTRA

Example *2. Let X be a continuous r.v. whose probability density function is given by

$$f(x) = \begin{cases} \alpha x + \beta, & 0 \leq x \leq 1 \\ 0, & otherwise \end{cases}$$

where α, β are constants.

(a) Find an equation that relates α and β.

(b) If $E(X) = \frac{5}{8}$, find the values of α and β. Hence, compute the mean and variance of $(2X^3 - 1)$.

EXTRA

Soln. (a) from the property $\int_{-\infty}^{\infty} f(x)dx = 1$, it follows that

$$\begin{aligned} \int_0^1 (\alpha x + \beta)dx = 1 &\Longrightarrow \left[\frac{\alpha}{2}x^2 + \beta x\right]_0^1 = 1 \\ &\Longrightarrow \alpha + 2\beta - 2 = 0. \end{aligned} \tag{1}$$

(b) $E(X) = 1$ gives

$$\begin{aligned} \int_0^1 x(\alpha x + \beta)dx = \frac{5}{8} &\Longrightarrow \left[\frac{\alpha x^3}{3} + \frac{\beta x^2}{2}\right]_0^1 = \frac{5}{8} \\ &\Longrightarrow 8\alpha + 12\beta - 15 = 0. \end{aligned} \tag{2}$$

EXTRA

Solving (1) and (2), we get $\alpha = \frac{3}{2}$, $\beta = \frac{1}{4}$. Now,

$$\begin{aligned} E(2X^3 - 1) &= 2E(X^3) - 1 \\ Var(2X^3 - 1) &= 2^2 Var(X^3) = 4Var(X^3) \\ &= 4\left\{E(X^6) - [E(X^3)]^2\right\}. \end{aligned} \tag{3}$$

Hence, we shall compute $E(X^3)$ and $E(X^6)$ as follows:

$$\begin{aligned} E(X^3) &= \int_0^1 x^3 \left(\tfrac{3x}{2} + \tfrac{1}{4}\right) dx = \int_0^1 \left(\tfrac{3}{2}x^4 + \tfrac{1}{4}x^3\right) dx \\ &= \left[\tfrac{3}{10}x^5 + \tfrac{1}{16}x^4\right]_0^1 = \tfrac{29}{80} \end{aligned}$$

EXTRA

$$\begin{aligned} E(X^6) &= \int_0^1 x^6 \left(\frac{3x}{2} + \frac{1}{4}\right) dx = \int_0^1 \left(\frac{3}{2}x^7 + \frac{1}{4}x^6\right) dx \\ &= \left[\frac{3}{16}x^8 + \frac{1}{28}x^7\right]_0^1 = \frac{25}{112} \end{aligned}$$

Substituting into (3), we get

$$E(2X^3 - 1) = 2E(X^3) - 1 = 2\left(\frac{29}{80}\right) - 1 = -\frac{11}{40}$$

$$\begin{aligned} Var(2X^3 - 1) &= 4\left\{E(X^6) - [E(X^3)]^2\right\} = 4\left\{\frac{25}{112} - \left(\frac{29}{80}\right)^2\right\} \\ &= 0.3672. \end{aligned}$$

EXTRA

Summary of $E(X)$, $Var(X)$

1. The *mean* of a probability distribution of a r.v. X is a measure of the centrality of the probability distribution. It is called the *expected value* of X, sometimes called the *expectation* of X.

$$\mu_X = E(X) = \begin{cases} \sum_x xP(x), & X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} xf(x)dx, & X \text{ is continuous.} \end{cases}$$

2. **General formula**

$$E[g(X)] = \begin{cases} \sum_x g(x)P(x), & X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} g(x)f(x)dx, & X \text{ is continuous.} \end{cases}$$

3. **The *variance* of a r.v. X is a measure of the spread of the possible values of the r.v. about the mean. It gives us an idea of the variation or uncertainty associated with the r.v: the larger the variance, the further away from the mean are**

possible values of the r.v.

$$\sigma_X^2 = Var(X) = E[(X - \mu_X)^2] = E(X^2) - [E(X)]^2$$

4. The *standard deviation* of a r.v. X is

$$\sigma_X = \sqrt{Var(X)}.$$

5. a, b are constants

$$E(aX+b) = aE(X)+b, \qquad Var(aX+b) = a^2 Var(X)$$

8. Normal Distribution

- The most important example of a *continuous* r.v. is the *normal* r.v., whose density function has a *symmetrical bell-shaped graph*. It has played a significant role in the study of random phenomenon in nature. Many naturally occurring random phenomena are approximately normal. For e.g.,

  1. the diameter of bolts
  2. weights of packages of food

3. heights of people

- A r.v. X is called a *normal r.v.* (or said to be *normally distributed*) if its probability density function is given by

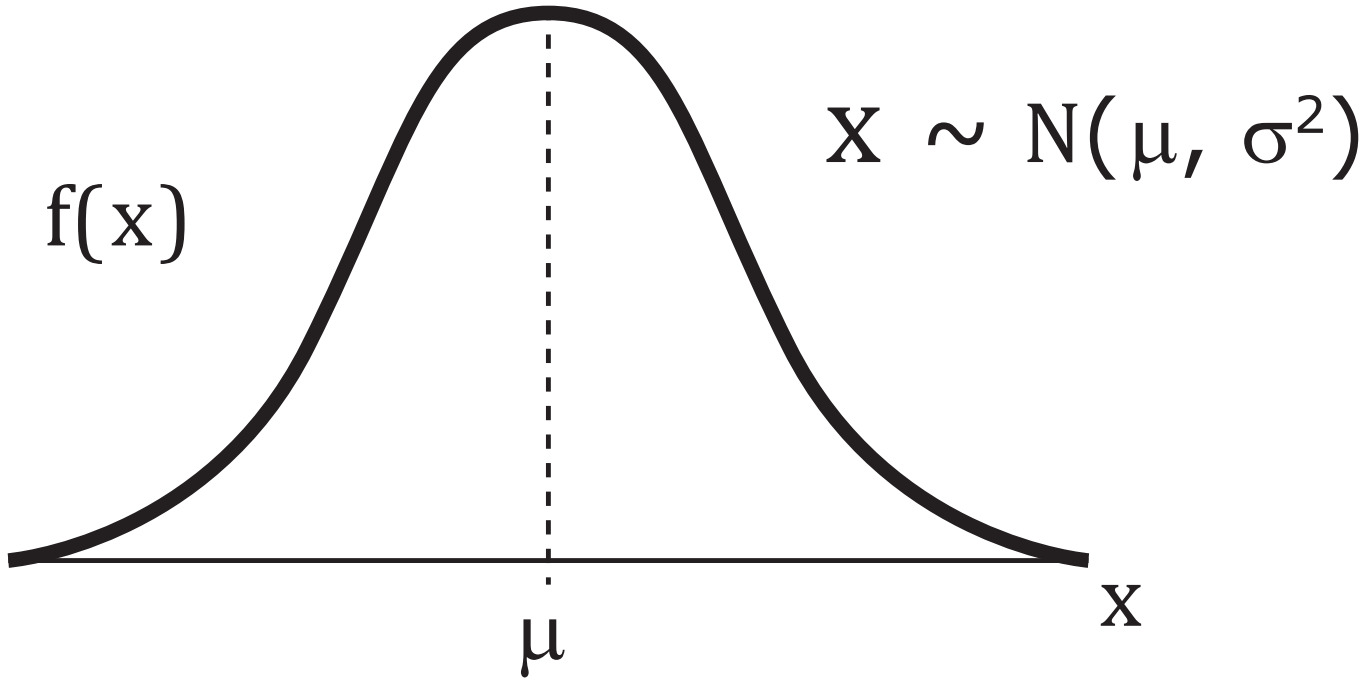
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \quad -\infty < x < \infty$$

where μ and σ^2 are respectively the mean and the variance of X.

We write

$$X \sim N(\mu, \sigma^2)$$

$\iff$ X **is normally distributed with mean** μ **and variance** σ^2**.**



- Let $X \sim N(\mu, \sigma^2)$. Define

$$Z = \frac{X - \mu}{\sigma}.$$

Then, Z has zero mean and unit variance (show it), and is normally distributed, i.e.,

$$Z \sim N(0, 1).$$

Z is called a *standard normal r.v.*

- **The distribution function of Z is**

$$\phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} \exp\left(-\frac{y^2}{2}\right) dy.$$

Values of $\phi(z)$ are given in Table A1.

8.1. Evaluating Standard Normal Probabilities

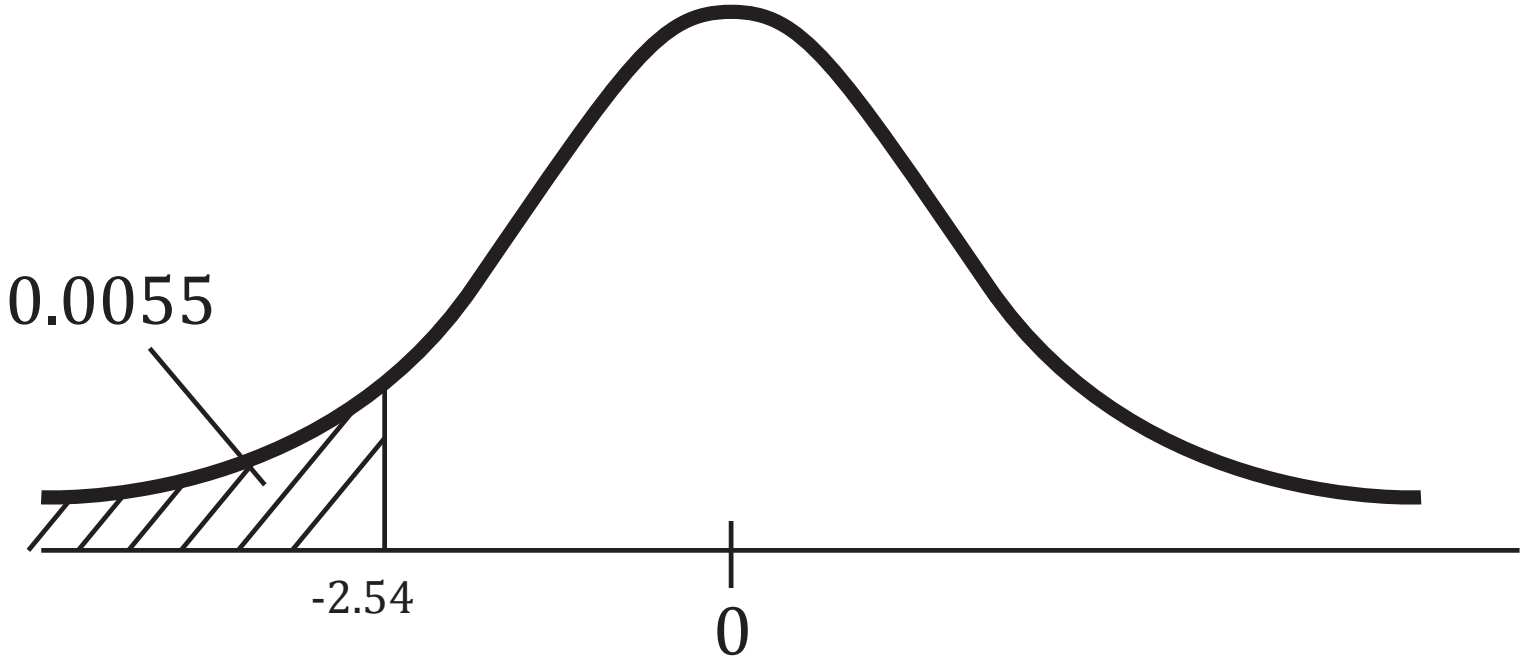
Table A1 gives the area under the standard normal curve between -3.49 and z, where $-3.49 \leq z \leq 3.49$ and in steps of 0.01. The shaded area is, as indicated by the picture in the table,

$$\phi(z) = P(Z \leq z).$$

Example *1. Find the following probabilities:

(a) $P(Z \leq -2.54)$

$$P(Z \leq -2.54) = \phi(-2.54) = 0.0055.$$

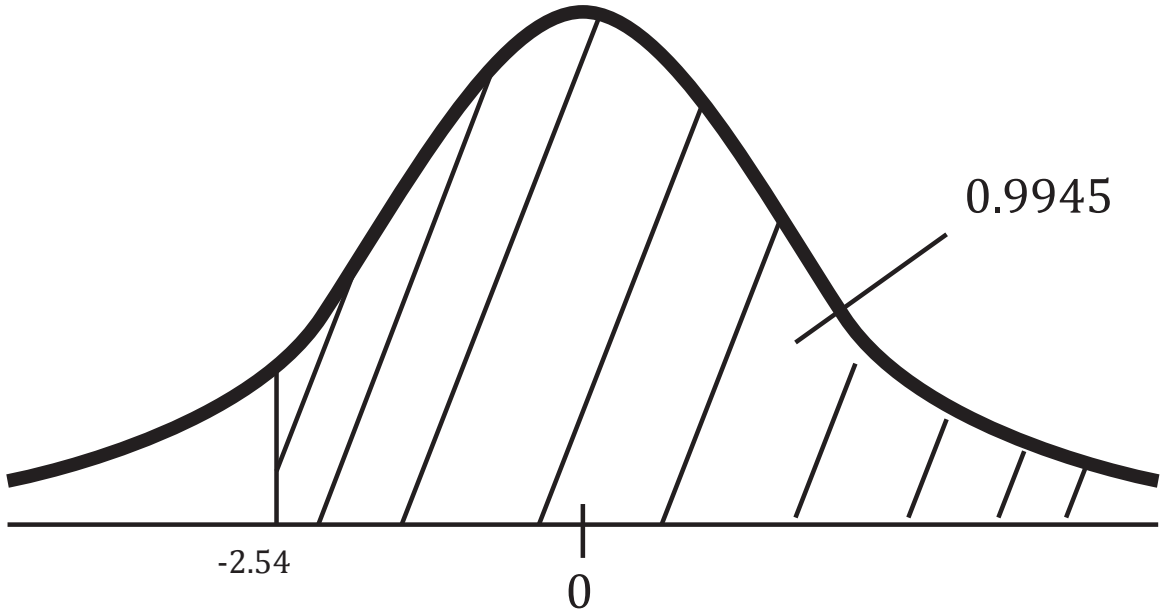


(b) $P(Z > -2.54)$

$$P(Z > -2.54) = 1 - P(Z \leq -2.54) = 1 - \phi(-2.54)$$
$$= 1 - 0.0055 = 0.9945.$$

Since Z is a continuous r.v.,

$$P(Z \geq -2.54) = P(Z > -2.54) = 0.9945.$$

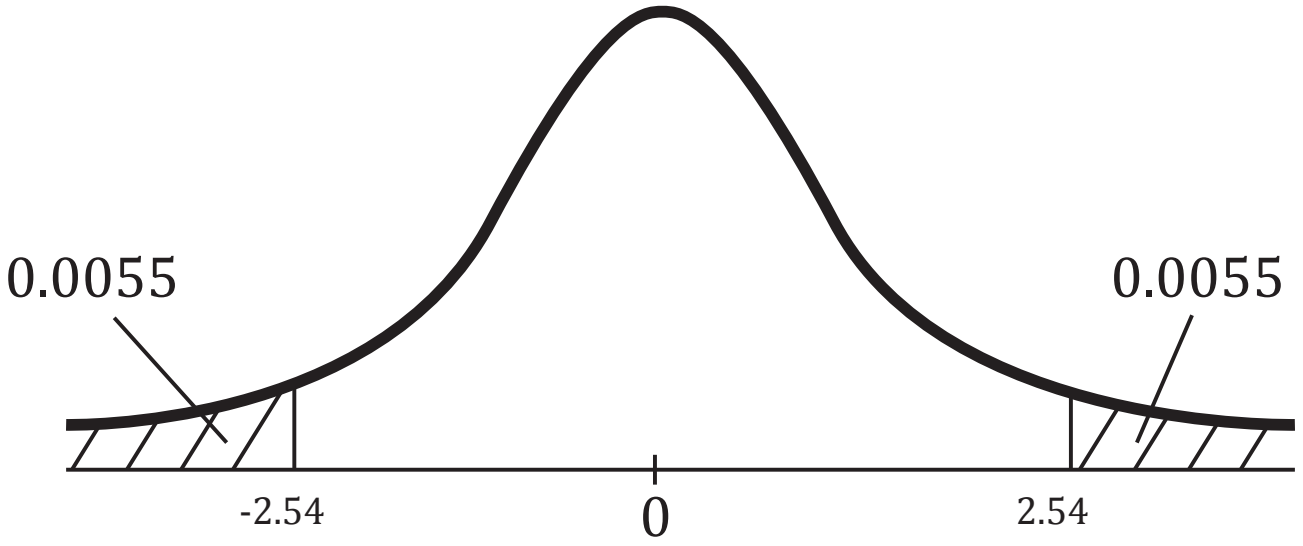


(c) $P(Z \geq 2.54)$

$$P(Z \geq 2.54) = 1 - P(Z < 2.54) = 1 - \phi(2.54)$$
$$= 1 - 0.9945 = 0.0055.$$

Note: by symmetry of the normal curve,

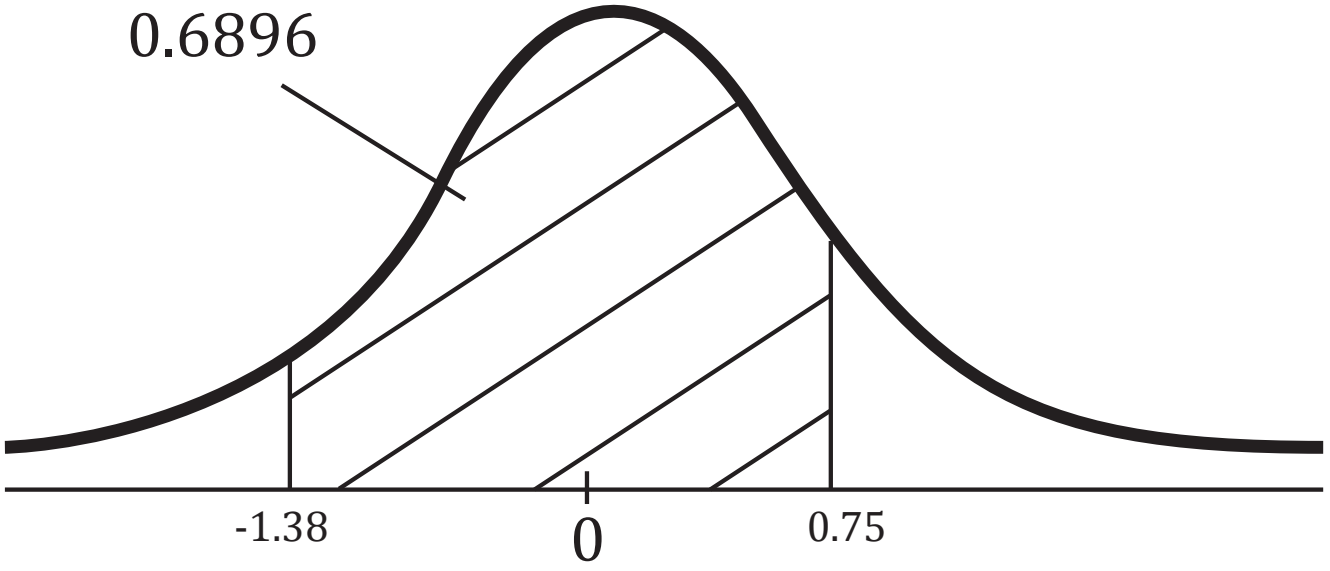
$$P(Z \geq 2.54) = P(Z \leq -2.54).$$



(d) $P(-1.38 \leq Z \leq 0.75)$

This is the area under the standard normal curve between -1.38 and 0.75.

$$\begin{aligned} P(-1.38 \leq Z \leq 0.75) &= \phi(0.75) - \phi(-1.38) \\ &= 0.7734 - 0.0838 = 0.6896. \end{aligned}$$



8.2. Evaluating Arbitrary Normal Probabilities

Suppose X is normally distributed, say, $X \sim N(\mu, \sigma^2)$.

We need to convert X to Z, the standard normal r.v.

$$Z = \frac{X - \mu}{\sigma}$$

$$\begin{aligned}
\textbf{e.g. } P(a \leq X \leq b) &= P\left(\frac{a-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) \\
&= P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right) \\
&= \phi\left(\frac{b-\mu}{\sigma}\right) - \phi\left(\frac{a-\mu}{\sigma}\right).
\end{aligned}$$

Example 8.2. Let $X \sim N(40, 2.25)$. Find $P(39 \leq X \leq 42)$.

Soln. We have $\mu = 40$ and $\sigma = \sqrt{2.25} = 1.5$. Therefore,

$$\begin{aligned} P(39 \leq X \leq 42) &= P\left(\frac{39-\mu}{\sigma} \leq Z \leq \frac{42-\mu}{\sigma}\right) \\ &= P(-0.67 \leq Z \leq 1.33) \\ &= \phi(1.33) - \phi(-0.67) \\ &= 0.9082 - 0.2514 = 0.6568. \end{aligned}$$

What is X ?

mean / variance / std deviation

of SOMETHING

Let X = this SOMETHING

SOMETHING has ... prob distribution

Let X = this SOMETHING

Example 8.3. A certain brand of battery lasts on the average 3 weeks, with a standard deviation of 0.5 week. Assume that the battery lives are normally distributed.

(a) Find the probability that a given battery will last less than 2.3 weeks.

(b) What percentages of batteries will last more than 4 weeks?

(c) Suppose at least 80% of batteries will last at least α weeks. Find the largest possible α.

Soln. Let $X =$ battery life. $X \sim N(3, 0.5^2)$.

Let $X =$ battery life. $X \sim N(3, 0.5^2)$.

(a) Find the probability that a given battery will last less than 2.3 weeks.

$$\begin{aligned} P(X < 2.3) &= P\left(Z < \tfrac{2.3-3}{0.5}\right) \\ &= P(Z < -1.4) = \phi(-1.4) = 0.0808 \end{aligned}$$

(b) What percentages of batteries will last more than 4 weeks?

Probability required is $P(X > 4)$.

$$\begin{aligned} P(X > 4) &= P\left(Z > \tfrac{4-3}{0.5}\right) \\ &= P(Z > 2) = 1 - P(Z \leq 2) = 1 - \phi(2) \\ &= 1 - 0.9772 = 0.0228 \end{aligned}$$

Hence, 2.28% of batteries will last more than 4 weeks.

(c) Suppose at least 80% of batteries will last at least α weeks. Find the largest possible α.

The statement means $P(X \geq \alpha) \geq 0.8$.

$$\begin{aligned} P(X \geq \alpha) &= P\left(Z \geq \tfrac{\alpha-3}{0.5}\right) \\ &= 1 - P\left(Z < \tfrac{\alpha-3}{0.5}\right) = 1 - \phi\left(\tfrac{\alpha-3}{0.5}\right) \end{aligned}$$

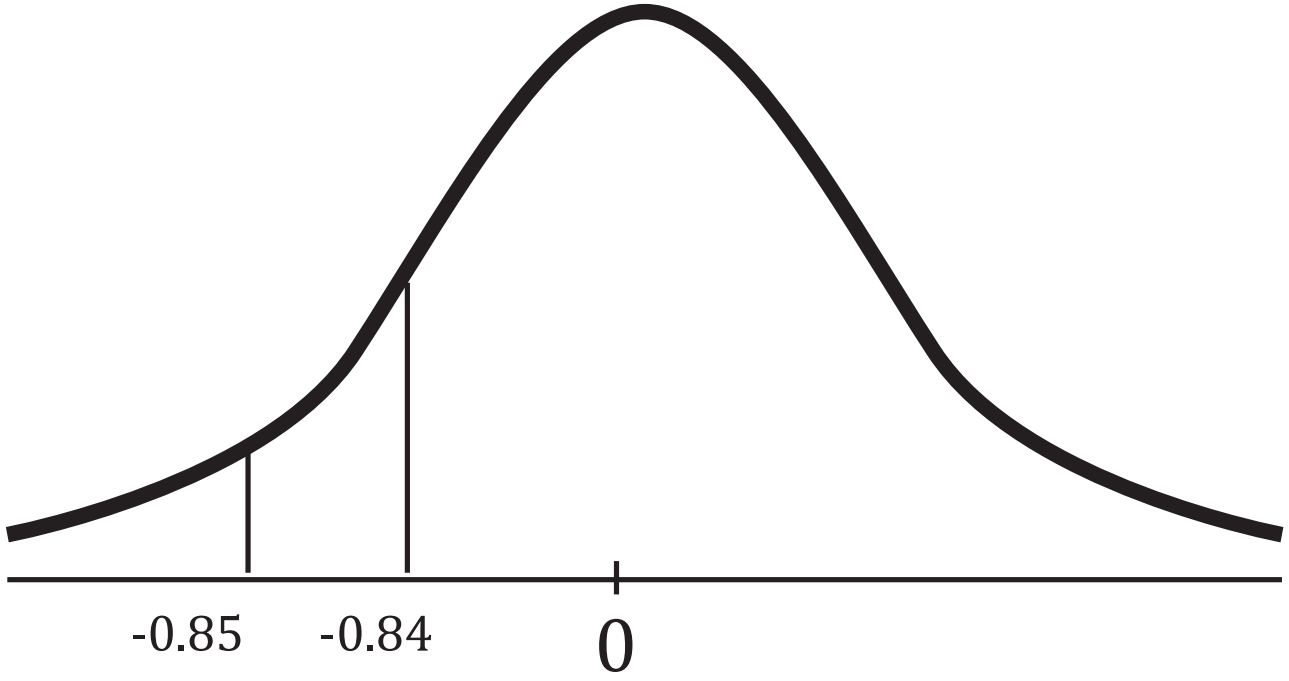
Now, $P(X \geq \alpha) \geq 0.8$ implies

$$1 - \phi\left(\frac{\alpha - 3}{0.5}\right) \geq 0.8$$

$$\Longrightarrow \phi\left(\frac{\alpha-3}{0.5}\right) \leq 0.2. \qquad (1)$$

From Table A1, we see that

$$\phi(-0.84) = 0.2005, \qquad \phi(-0.85) = 0.1977.$$



Noting (1), set

$$\frac{\alpha - 3}{0.5} \leq -0.85 \Rightarrow \alpha \leq -0.85(0.5) + 3 = 2.575.$$

Hence, the largest possible $\alpha = 2.575$ weeks.

Example *2. The weight of a fishball made in a factory is normally distributed. Its mean value is 15g with a standard deviation of 0.2g.

(a) Find the probability that a randomly selected fishball has a weight of less than 14.6g.

(b) A fishball fails the quality test if its weight does NOT fall between 14.75g to 15.25g. What is the passing rate of a sufficiently large batch of fishballs?

(c) To control cost, the factory manager stipulates that at least 75% of fishballs made should weigh less than β grams. Find

EXTRA

the smallest possible β.

Soln. Let $X =$ the weight of a fishball made in the factory. $X \sim N(15, 0.2^2)$.

(a) Find the probability that a randomly selected fishball has a weight of less than 14.6g.

$$P(X < 14.6) = P\left(\frac{X-15}{0.2} < \frac{14.6-15}{0.2}\right)$$

$$= P(Z < -2) = 0.0228$$

EXTRA

(b) A fishball fails the quality test if its weight does NOT fall between 14.75g to 15.25g. What is the passing rate of a sufficiently large batch of fishballs?

The passing rate is just the probability $P(14.75 \leq X \leq 15.25)$.

$$\begin{aligned} P(14.75 \leq X \leq 15.25) &= P(\tfrac{14.75-15}{0.2} \leq \tfrac{X-15}{0.2} \leq \tfrac{15.25-15}{0.2}) \\ &= P(-1.25 \leq Z \leq 1.25) \\ &= \phi(1.25) - \phi(-1.25) = 0.7888 \end{aligned}$$

The passing rate is 78.88%.

EXTRA

(c) To control cost, the factory manager stipulates that at least 75% of fishballs made should weigh less than β grams. Find the smallest possible β.

The statement means $P(X < \beta) \geq 0.75$.

Note that $P(X < \beta) = P\left(Z < \frac{\beta - 15}{0.2}\right) = \phi\left(\frac{\beta - 15}{0.2}\right)$. So,

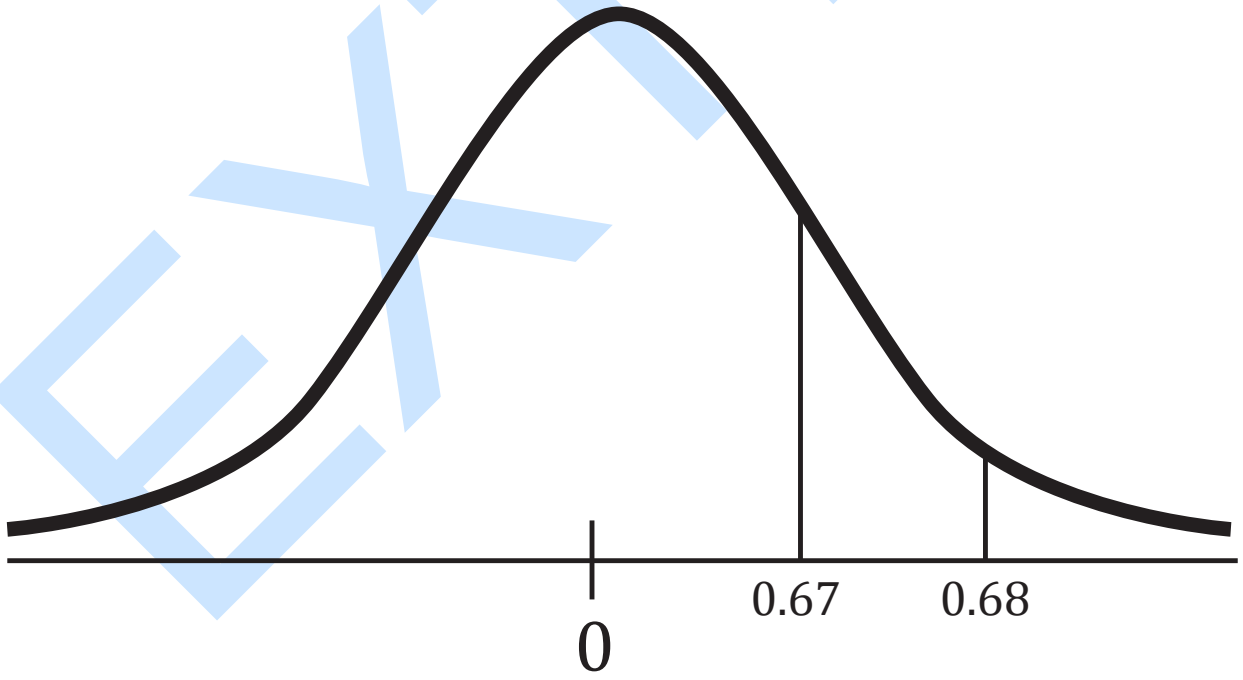
$$P(X < \beta) \geq 0.75 \Longrightarrow P\left(Z < \frac{\beta - 15}{0.2}\right) \geq 0.75$$

EXTRA

$$\Longrightarrow \phi\left(\frac{\beta - 15}{0.2}\right) \geq 0.75 \qquad (2)$$

From Table A1, we have

$$\phi(0.68) = 0.7517, \qquad \phi(0.67) = 0.7486.$$



EXTRA

Due to (2), set

$$\frac{\beta - 15}{0.2} \geq 0.68 \Rightarrow \beta \geq 0.68(0.2) + 15 = 15.136.$$

Hence, the smallest β is 15.136.

EXTRA

Summary

1. The *normal distribution* is a symmetrical bell-shaped curve defined by two parameters - its mean, μ, and standard deviation, σ. $X \sim N(\mu, \sigma^2)$.

2. The *standard normal distribution* is a continuous, symmetrical, bell-shaped distribution that has a mean of 0 and a standard deviation of 1. $Z \sim N(0, 1)$.

3. If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.

9. Poisson Distribution

- A type of distribution useful in describing the no. of events that will occur in a specific period of time, distance or space is the *Poisson distribution*.

- The *Poisson distribution* models the no. of random arrivals of events within a period of time or space.

- Poisson distribution is a *discrete* probability distribution.

- Here are some examples of experiments for which the r.v.

X can be modelled by the Poisson r.v. :

1. **The no. of telephone calls received by a switchboard during a given period of time.**
2. **The no. of customers entering a bank during lunch time.**
3. **The no. of machine breakdowns during a given day.**
4. **The no. of claims against an insurance company during a given week.**
5. **The no. of flaws on a 1-sq m piece of marble.**

Memorable examples of Poisson r.v. :

$X =$ **the no. of girlfreinds a guy has from 8 to 80 years old**

$X =$ **the no. of pimples on a face**

In each example, X **represents the no. of events occurring in a period of time or space in which an** $average$ **of** λ **such events can be expected to occur.**

- The only assumptions needed when one uses the Poisson distribution to model experiments are that counts or events occur *randomly* and *independently of one another*.

- A r.v. X is called a *Poisson r.v.* with parameter λ (the mean rate of the event occurrence) if its probability mass function is given by

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \cdots, \infty.$$

  We write $X \sim P(\lambda)$.

- **The distribution function of X is**

$$F(x) = P(X \leq x) = e^{-\lambda} \sum_{k \leq x} \frac{\lambda^k}{k!}.$$

- **The mean and variance of the Poisson r.v. X are (show it)**

$$\mu_X = \lambda \qquad and \qquad \sigma_X^2 = \lambda.$$

Note: λ represents both the mean and the variance of the Poisson distribution.

What is X ?

mean / variance / std deviation of SOMETHING

Let X = this SOMETHING

SOMETHING has ... prob distribution

Let X = this SOMETHING

Example 9.1. The no. of customers arriving at a shop during any 1-hour period is known to be a Poisson r.v. with $\lambda = 2$.

(a) What is the probability that no customers will arrive during any 1-hour period?

(b) What is the probability that at most 1 customer will arrive during any 1-hour period?

(c) Find the probability that more than 3 customers will arrive during any 1-hour period.

Soln. Let $X =$ the no. of customers arriving at a shop during any 1-hour period. $X \sim P(2)$. The pmf of X is

$$P(X = k) = \frac{e^{-2}2^k}{k!}, \qquad k = 0, 1, 2, \cdots, \infty.$$

(a) What is the probability that no customers will arrive during any 1-hour period?

The probability required is

$$P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2} = 0.135.$$

(b) What is the probability that at most 1 customer will arrive during any 1-hour period?

The probability required is

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$= \frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} = e^{-2}(1 + 2) = 0.406.$$

(c) Find the probability that more than 3 customers will arrive during any 1-hour period.

The probability required is

$$P(X > 3)$$

$$= 1 - P(X \leq 3)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$$

$$= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right]$$

$$= 1 - e^{-2}\left[1 + 2 + \frac{4}{2} + \frac{8}{6}\right] = 0.143.$$

Example 9.2. At a housing estate, power failures occur randomly and independently of one another at the rate of 4.2 per year. Thus, the probability distribution of X, the number of power failures occurring during a given period of time, can be modelled by the Poisson distribution. Find the following probabilities.

(a) No power failures during a given 4-month period.

(b) No more than 2 power failures occur in a given 6-month period.

Soln.

(a) No power failures during a given 4-month period.

Since power failures occur at the rate of 4.2 per year, so we expect the no. of power failures during any 4-month period to be

$$\lambda = \frac{4.2}{3} = 1.4.$$

Let X = the no. of power failures during any 4-month period. $X \sim P(1.4)$.

The probability required is

$$P(X = 0) = \frac{e^{-1.4}(1.4)^0}{0!} = e^{-1.4} = 0.247.$$

(b) No more than 2 power failures occur in a given 6-month period.

Since power failures occur at the rate of 4.2 per year, so we expect the no. of power failures during any 6-month period to be

$$\lambda = \frac{4.2}{2} = 2.1.$$

Let X = the no. of power failures during any 6-month period. $X \sim P(2.1)$.

The probability required is

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= \frac{e^{-2.1}(2.1)^0}{0!} + \frac{e^{-2.1}(2.1)^1}{1!} + \frac{e^{-2.1}(2.1)^2}{2!} \\ &= e^{-2.1}\left[1 + 2.1 + \frac{(2.1)^2}{2}\right] = 0.650. \end{aligned}$$

Example *1. An insurance company has on the average receives 5 car accident claims per day.

(a) Find the probability that the company receives at least 2 car accident claims on a randomly selected day.

(b) One day there are 4 car accident claims reported by noon. What is the probability that there are at most 6 car accident claims for that day?

Soln. Let $X =$ the no. of car accident claims that the insurance company receives in a day. We assume X is a

EXTRA

Poisson r.v. with $\lambda = 5$, i.e., $X \sim P(5)$.

(a) Find the probability that the company receives at least 2 car accident claims on a randomly selected day.

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - (\frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!}) \\ &= 0.9596 \end{aligned}$$

EXTRA

(b) One day there are 4 car accident claims reported by noon. What is the probability that there are at most 6 car accident claims for that day?

EXTRA

$$P(X \leq 6 \mid X \geq 4) = \frac{P(4 \leq X \leq 6)}{P(X \geq 4)}$$

$$= \frac{P(X=4)+P(X=5)+P(X=6)}{1-P(X<4)}$$

$$= \frac{P(X=4)+P(X=5)+P(X=6)}{1-[P(X=0)+P(X=1)+P(X=2)+P(X=3)]}$$

$$= \frac{0.4972}{0.7350} = 0.6765$$

EXTRA

Example *2. In the previous example, suppose it is known that within the first week of August the insurance company has already received n car accident claims. What is the probability that there are at least m $(m > n)$ car accident claims for the month of August?

Soln. Let $X =$ the no. of car accident claims that the insurance company receives in August.

$$X \sim P(31 * 5) = P(155)$$

EXTRA

$$P(X \geq m \mid X \geq n)$$

$$= \frac{P(X \geq m)}{P(X \geq n)}$$

$$= \frac{1 - P(X < m)}{1 - P(X < n)}$$

$$= \frac{1 - [P(X = 0) + P(X = 1) + \cdots + P(X = m - 1)]}{1 - [P(X = 0) + P(X = 1) + \cdots + P(X = n - 1)]}$$

EXTRA

Summary

1. The *Poisson probability distribution* is a model for finding the probability of count data resulting from any experiment, where the count represents the no. of rare events observed in a given period of time or space.

   X = the no. of girlfreinds a guy has from 8 to 80 years old

   X = the no. of pimples on a face

2. $X \sim P(\lambda)$, where λ is the mean rate of the event occurrence

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \cdots, \infty.$$

3. $E(X) = Var(X) = \lambda$

10. Binomial Distribution

10.1. Binomial Experiment

A *binomial experiment* is an experiment that possesses the following properties:

1. **The experiment consists of n identical trials.**

2. **Each trial results in one of two outcomes. We will call one outcome 'success', and the other 'failure'.**

3. The *probability of success* on a single trial is equal to p and remains the same from trial to trial. The *probability of failure* is equal to $(1-p)$.

4. The trials are *independent*, i.e., the probability of success in a trial is not dependent on the outcome of any other trial.

5. The experimenter is interested in X, the no. of successes observed during n trials.

Binomial

Very few real life situations will completely satisfy the requirements for a binomial experiment, but this is of little consequence as long as the lack of agreement is moderate and does not affect the end result.

The following experiments often exhibit, to a reasonable degree of approximation, the characteristics of a binomial experiment.

1. **A single missile results in a successful or unsuccessful launching.**

2. **A new drug is effective or ineffective when administered to a single patient.**

3. **A manufactured item selected from a production line is defective or non-defective.**

4. **With each customer, either a salesman will consummate a sale or no sale will result.**

The binomial experiment is an excellent model for many sampling situations particularly market surveys that result in 'yes' or 'no' types of data. As long as the sample size n

is small relative to the population size, then the binomial experiment is an excellent model. A common guideline is that the sample size n should be no more than 5% of the population size.

10.2. Binomial r.v.

- A *binomial r.v.* X is associated with some experiment in which n independent trials are performed and X represents the no. of successes that occur in n trials. The probability of success on a single trial is equal to p.

X = the no. of successes that occur in n trials of a binomial experiment

e.g.

X = the no. of good apples in a box of 100 apples.

X = the no. of bad apples in a box of 100 apples.

- X is *discrete*.

- X is a *binomial r.v.* with parameters (n, p) if its probability mass function is given by

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \cdots, n$$

where $0 \leq p \leq 1$ and

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

is the *binomial coefficient*. We write

$$X \sim B(n, p).$$

- The distribution function of X is

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}.$$

- The mean and variance of the binomial r.v. X are (show it)

$$\mu_X = np \qquad and \qquad \sigma_X^2 = np(1-p).$$

Example 10.1. In a durian-tree orchard, it is found that 5% of the durians produced are rejected because they are tasteless. What is the probability that a sample of 12 durians contains (a) exactly 2, (b) not more than 2, (c) at least 2 rejects?

– – – – – – – – – – – – – – – – – – – –

$X =$ the no. of successes that occur in n trials of a binomial experiment

$X =$ the no. of rejected durians in a sample of 12 durians

Soln. Let $X =$ the no. of rejects in a sample of 12 durians. The fraction of rejected durians is $p = 5\% = 0.05$. Hence, $X \sim B(12, 0.05)$.

$$P(X = k) = \binom{12}{k}(0.05)^k(1-0.05)^{12-k}, \quad k = 0, 1, \cdots, 12.$$

(a) exactly 2 rejects

$$P(X = 2) = \binom{12}{2}(0.05)^2(0.95)^{12-2} = 0.099$$

(b) not more than 2 rejects

$$P(X \leq 2)$$

$$= P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \binom{12}{0}(0.05)^0(0.95)^{12-0} + \binom{12}{1}(0.05)^1(0.95)^{12-1}$$
$$+ \binom{12}{2}(0.05)^2(0.95)^{12-2}$$

$$= 0.54 + 0.341 + 0.099 = 0.98$$

(c) at least 2 rejects

$$
\begin{aligned}
P(X \geq 2) &= 1 - P(X < 2) \\
&= 1 - [P(X = 0) + P(X = 1)] \\
&= 1 - (0.54 + 0.341) = 0.12
\end{aligned}
$$

Example *1. Consider a digital communication system in which 1's and 0's are transmitted. The probability of an error (i.e., a 1 received as a 0, or vice versa) is 0.001. What is the probability of (a) less than 2, (b) more than 5 errors in the transmission of 50 bits? (A bit is a 1 or a 0)

– – – – – – – – – – – – – – – – – –

$X =$ the no. of successes that occur in n trials of a binomial experiment

$X =$ the no. of errors in the transmission of 50 bits

EXTRA

Soln. Assume that the occurrence of an error in a particular bit does not influence the probability of occurrence of an error in any other bit. Then,

$$X = \text{the no. of errors in 50 bits}$$

is a binomial r.v. with $(n, p) = (50, 0.001)$.

$$P(X = k) = \binom{50}{k}(0.001)^k(1-0.001)^{50-k}, \quad k = 0, 1, \cdots, 50$$

EXTRA

(a) The probability of less than 2 errors in the transmission of 50 bits is

$$\begin{aligned} P(X < 2) &= P(X = 0) + P(X = 1) \\ &= 0.999^{50} + \binom{50}{1}(0.001)(0.999)^{49} \\ &= 0.9988 \end{aligned}$$

EXTRA

(b) The probability of more than 5 errors in the transmission of 50 bits is

$$
\begin{aligned}
& P(X > 5) \\
&= 1 - P(X \leq 5) \\
&= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\
&\qquad + P(X = 4) + P(X = 5)] \\
&= 1 - \Big[0.999^{50} + \binom{50}{1}(0.001)(0.999)^{49} + \binom{50}{2}(0.001)^2(0.999)^{48} \\
&\qquad + \binom{50}{3}(0.001)^3(0.999)^{47} + \binom{50}{4}(0.001)^4(0.999)^{46} \\
&\qquad + \binom{50}{5}(0.001)^5(0.999)^{45}\Big]
\end{aligned}
$$

EXTRA

Summary

1. The *binomial probability distribution* gives the probability of the no. of successes observed during n identical independent trials of a binomial experiment. The binomial experiment is an excellent model for many sampling situations, particularly market surveys that result in 'yes' or 'no' types of data.

2. $X \sim B(n, p)$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \cdots, n$$

3. $E(X) = np, \quad Var(X) = np(1-p)$

10.4. Normal Approximation of Binomial Distribution

- The binomial probabilities become increasingly difficult to compute as n gets larger. However, there is a way to approximate binomial distribution by means of a normal distribution when an exact computation is impractical.

  E.g. $X \sim B(1000, 0.05)$, find $P(X \geq 500)$.

$$\begin{aligned} P(X \geq 500) &= 1 - P(X < 500) \\ &= 1 - [P(X = 0) + \cdots + P(X = 499)] \end{aligned}$$

- As the binomial distribution is *discrete* and the normal distribution is *continuous*, a suitable modification must be applied. Note that the discrete r.v. X has no values between k and $k+1$. Rather, its probabilities are concentrated at the integers $0, 1, 2, \cdots, n$. However, the continuous normal r.v. has zero probabilities at these integers; it has non-zero probability only at intervals. Therefore, to use the normal distribution to approximate binomial probabilities, intervals of ± 0.5 must be used.

- **Let $X \sim B(n, p)$. If $np \geq 5$ and $n(1-p) \geq 5$, the binomial r.v. can be approximated by the normal r.v. X with mean np and variance $np(1-p)$.**

$$X \sim B(n, p) \text{ with } np \geq 5 \text{ and } n(1-p) \geq 5$$
$$\Rightarrow \quad X \text{ is approximately } N(np, np(1-p)).$$

- The **binomial probability**

$$P_b(X = k)$$

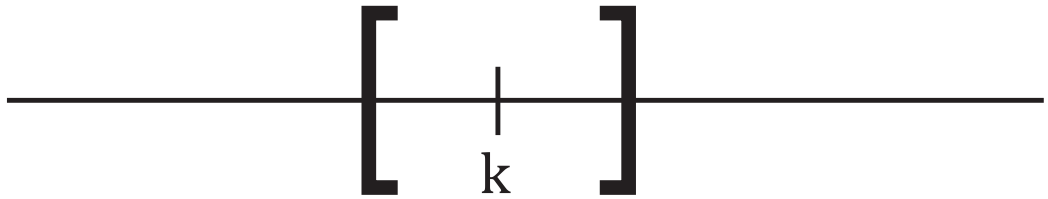
where $X \sim B(n, p)$, can be approximated by the **normal probability**

$$P_n(k - 0.5 \leq X \leq k + 0.5)$$

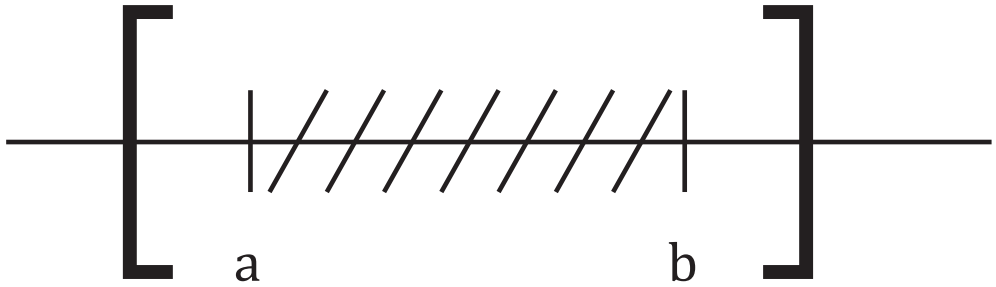
where $X \sim N(np, np(1 - p))$. The value of 0.5 is the *continuity correction factor.*

Envelope principle

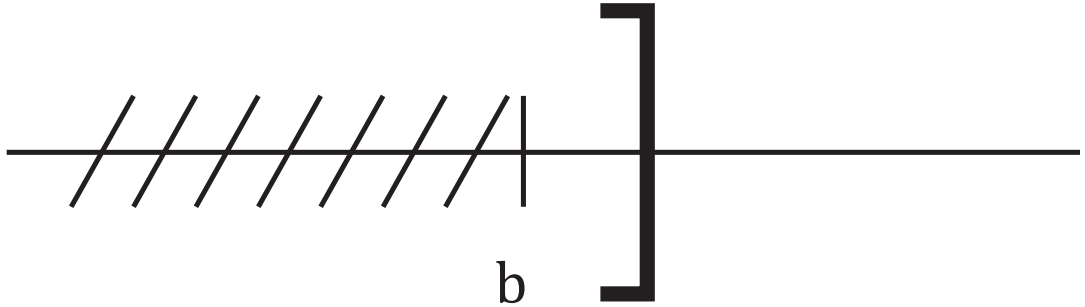
$$P_b(X = k) \approx P_n(k - 0.5 \leq X \leq k + 0.5)$$



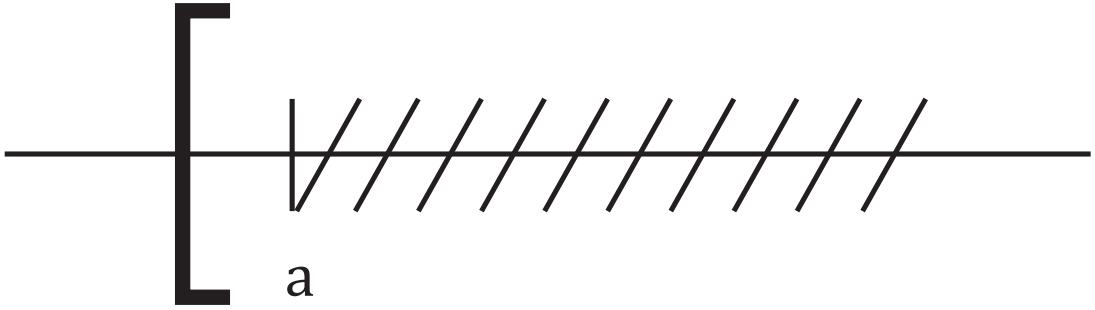
$$P_b(a \leq X \leq b) \approx P_n(a - 0.5 \leq X \leq b + 0.5)$$



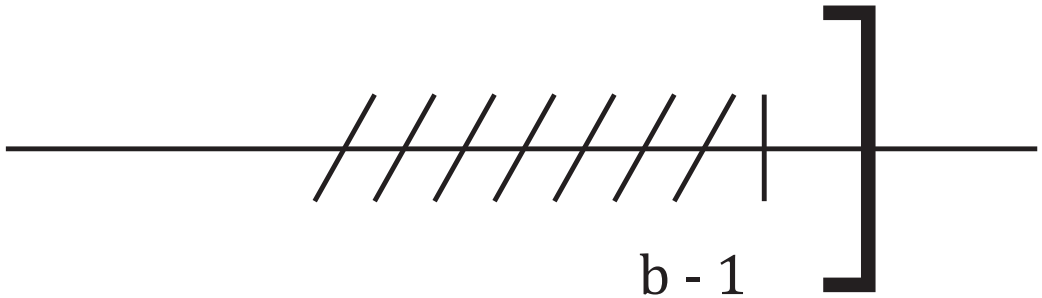
$$P_b(X \leq b) \approx P_n(X \leq b + 0.5)$$



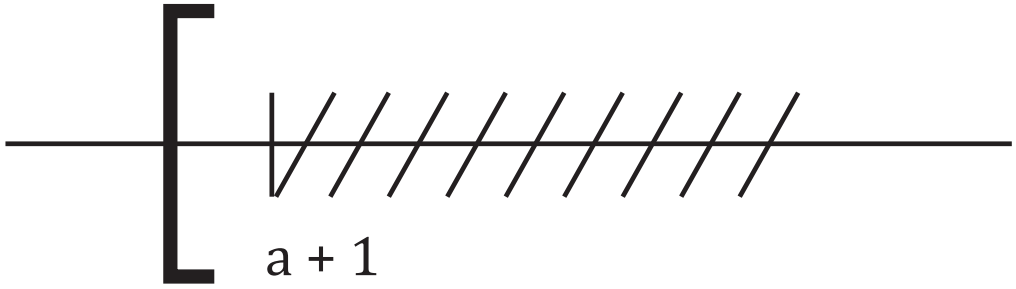
$$P_b(X \geq a) \approx P_n(X \geq a - 0.5)$$



$$P_b(X < b) = P_b(X \leq b - 1) \approx P_n(X \leq b - 1 + 0.5)$$



$$P_b(X > a) = P_b(X \geq a + 1) \approx P_n(X \geq a + 1 - 0.5)$$



Example 10.3. A fair coin is tossed 100 times. Find the probability that tails occur (a) exactly 60 times, (b) between 48 and 53 times inclusive, (c) less than 45 times.

– – – – – – – – – – – – – – – – – –

$X =$ the no. of successes that occur in n trials of a binomial experiment

$X =$ the no. of tails in 100 tosses

Soln. Let $X =$ the no. of tails in 100 tosses. X is a binomial r.v. with $n = 100,\ p = 0.5$. Since

$$np = 50 > 5 \qquad \text{and} \qquad n(1-p) = 50 > 5,$$

we can use the normal distribution to approximate the binomial probability.

We have

$$\mu = np = 50, \qquad \sigma = \sqrt{np(1-p)} = 5.$$

So the binomial r.v. $X \sim B(n, p)$ can be approximated by $X \sim N(50, 5^2)$.

(a)

$$
\begin{aligned}
P_b(X = 60) &\approx P_n(60 - 0.5 \leq X \leq 60 + 0.5) \\
&= P_n(59.5 \leq X \leq 60.5) \quad [X \sim N(50, 5^2)] \\
&= P_n\left(\frac{59.5 - 50}{5} \leq Z \leq \frac{60.5 - 50}{5}\right) \\
&= P_n(1.9 \leq Z \leq 2.1) \\
&= \phi(2.1) - \phi(1.9) = 0.9821 - 0.9713 = 0.0108
\end{aligned}
$$

The result agrees with the exact value of $P_b(X = 60)$ to 4 decimal places,

$$P_b(X = 60) = \binom{100}{60}(0.5)^{60}(0.5)^{100-60} = 0.0108.$$

(b)

$$P_b(48 \leq X \leq 53)$$

$$\approx P_n(48 - 0.5 \leq X \leq 53 + 0.5)$$

$$= P_n(47.5 \leq X \leq 53.5) \quad [X \sim N(50, 5^2)]$$

$$= P_n\left(\frac{47.5 - 50}{5} \leq Z \leq \frac{53.5 - 50}{5}\right)$$

Ans: 0.4495

(c)

$$P_b(X < 45) = P_b(X \leq 44)$$

$$\approx P_n(X \leq 44 + 0.5)$$

$$= P_n(X \leq 44.5) \quad [X \sim N(50, 5^2)]$$

$$= P_n\left(Z \leq \frac{44.5 - 50}{5}\right)$$

Ans: 0.1357

Summary

1.

$$X \sim B(n, p) \text{ with } np \geq 5 \text{ and } n(1-p) \geq 5$$

$$\Rightarrow X \text{ is approximately } N(np, np(1-p)).$$

2. Envelope principle